The First Retirement Age: Civil War Pensions and the Labor Supply Response to Age-Based Eligibility*

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Abstract

The 1907 Service and Age Pension Act granted automatic pension eligibility to Union veterans at age 62. Using the Costa Union Army dataset—21,302 veterans with linked 1900–1910 census records and observed pension histories—I estimate a regression discontinuity at the age-62 threshold. The first stage is clear: pension receipt under the 1907 Act jumps 10 percentage points at the cutoff. The panel RDD, which differences out permanent individual characteristics, yields a 7 percentage point decline in labor force participation at wider bandwidths, though the estimate at the optimal bandwidth is imprecise. The minimum detectable effect falls from 30 percentage points in prior work to 14 percentage points here. Observed pension records, panel structure, and health data from surgeons' certificates advance this from an underpowered exercise to a substantially more informative evaluation of America's first age-based social insurance program.

JEL Codes: H55, J26, N31, I38

Keywords: Civil War pensions, retirement, labor supply, regression discontinuity, panel data, Costa Union Army dataset

*This paper is the third revision of APEP Working Paper 0442 (available at https://github.com/SocialCatalystLab/ape-papers/tree/main/apep_0442). This revision replaces the IPUMS census data with the Costa Union Army dataset, which provides linked panel data across the 1900 and 1910 censuses, observed pension records, and surgeons' health certificates—enabling a first-stage analysis, panel RDD, and mechanism decomposition that were impossible with census cross-sections alone.

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1. Introduction

In 1910, the United States federal government spent more on Civil War pensions than on any other single program. The pension system consumed 28 percent of all federal expenditures—a share that modern Social Security, despite its vastly larger scale, has never matched (**?**). Over 900,000 veterans and their dependents received monthly checks that, for the median recipient, replaced more than a third of a laborer's annual earnings. This was America's first experiment with mass social insurance, predating the New Deal by a generation. Yet no study has brought adequate data to bear on the most fundamental question: did pension eligibility at a fixed age cause men to stop working?

Three years before the 1910 census was enumerated, the Service and Age Pension Act of 1907 drew a bright statutory line. Any Union veteran who had served ninety or more days and reached age 62 became automatically eligible for a monthly pension of $12, regardless of disability or prior application status. The pension rose to $15 at age 70 and $20 at age 75. For an unskilled laborer earning roughly $400 per year, the $144 annual pension represented a 36 percent income supplement—a transfer large enough to meaningfully alter the calculus of whether to continue working.

This paper exploits the age-62 threshold in a regression discontinuity design, using data that transforms what was previously an underpowered exercise into a substantially more informative study. The Costa Union Army dataset, assembled by Dora Costa and colleagues and archived at the NBER, links the military service records, pension files, surgeons' health certificates, and census records of approximately 39,000 white Union Army veterans (**??**). From this extraordinary resource I construct a sample of 21,302 veterans observed in the 1910 census, of whom 20,651 are also observed in the 1900 census—creating a ten-year panel that tracks the same individuals across time. Crucially, the dataset records *actual pension receipt*: whether each veteran received a pension, under which law (including the 1907 Act specifically), and the monthly dollar amount. For the first time, I can observe the first stage—the jump in pension take-up at the age-62 threshold—rather than merely assuming it.

The results tell a nuanced story. The first stage is clear and precisely estimated: pension receipt under the 1907 Act jumps by 10.2 percentage points at the age-62 cutoff ($p = 0.040$), and any pension receipt increases by 33.2 percentage points ($p = 0.004$). The jump is attenuated relative to a hypothetical zero-to-pension transition because many veterans below 62 already received disability-based pensions under earlier legislation—mean pension receipt is 29.2 percent even below the threshold.

The reduced-form effect on labor force participation is suggestive but bandwidth-sensitive. The cross-sectional RDD at the MSE-optimal bandwidth of 2.8 years yields a small, statistically

insignificant estimate (coefficient = 0.025, $p = 0.625$). The panel RDD—which differences 1910 and 1900 labor force participation for the same individual, absorbing all time-invariant confounders—produces a larger estimate of -0.071 at the optimal bandwidth ($p = 0.165$), which strengthens at wider bandwidths: -0.065 ($p = 0.060$) at a bandwidth of 5 years, and -0.075 ($p = 0.049$) at 8 years. The fuzzy RDD LATE, instrumenting pension receipt with the age threshold, is 0.339 but too imprecise for inference ($p = 0.460$), reflecting the modest 10 percentage point first stage.

The pre-treatment falsification test—whether 1900 labor force participation is smooth at the age-62-in-1907 cutoff—yields a marginally significant coefficient of 0.113 ($p = 0.067$). This is a real concern that I discuss honestly: it suggests possible composition differences across the cutoff that could contaminate the reduced-form estimates. The panel RDD partially addresses this by differencing out level differences, but does not eliminate concerns about differential trends.

The paper makes three contributions. First, I document for the first time the observed first stage of the 1907 Act's age-62 threshold. The 10 percentage point jump in pension receipt, combined with a $1.91 increase in monthly pension amounts, establishes that the threshold had real economic consequences—even in a setting where most veterans already received some form of pension. The first stage finding is itself substantively important: it reveals that 70 percent of veterans below age 62 did *not* receive pensions under the 1907 Act, meaning the age threshold created genuinely new income for a substantial minority.

Second, the panel RDD—differencing the same individual's labor force participation across the 1900 and 1910 censuses—absorbs all permanent individual characteristics that might differ across the cutoff. This is a material advance over cross-sectional designs, which must assume that veterans born in 1847 and 1849 are identical in all respects except pension eligibility. The panel design requires only that *changes* in outcomes are comparable, a weaker and more plausible assumption.

Third, the Costa dataset's health records from surgeons' certificates and detailed occupation codes enable mechanism analysis that was impossible with census data alone. I examine whether health deterioration, rather than pension income, drives retirement at the threshold; whether different occupational groups respond differently; and whether the pension effect operates through discrete exit from the labor force or gradual reduction in work intensity.

The minimum detectable effect at the optimal bandwidth falls from approximately 30 percentage points in the prior IPUMS-based analysis to 14.3 percentage points here—and to 11 percentage points at a bandwidth of 5 years. This tenfold increase in sample size and the addition of panel structure, observed pensions, and health data represent a qualitative transformation of the analysis. We can now rule out the massive labor supply elasticities

suggested by cross-sectional studies (**?**), while the consistent -0.065 to -0.075 panel estimates at wider bandwidths provide suggestive evidence that pension eligibility reduced labor force participation by about 7 percentage points—a moderate effect consistent with the attenuated first stage.

The paper proceeds as follows. Section 2 provides historical background on the Civil War pension system and the 1907 Act. Section 3 reviews the related literature. Section 4 develops the conceptual framework. Section 5 describes the Costa Union Army dataset and sample construction. Section 6 presents the empirical strategy. Section 7 reports the main results. Section 8 presents robustness checks. Section 9 explores extensions including subgroup heterogeneity, multi-cutoff dose-response analysis, occupation transitions, and health mechanisms. Section 10 discusses the findings and concludes.

2. Historical Background

2.1 The Civil War Pension System

The Union pension system began modestly in 1862 as disability compensation for soldiers injured in service. Over the following decades, political pressure from the Grand Army of the Republic—the organized veterans' lobby—and the electoral calculus of the Republican Party transformed it into a comprehensive old-age support system for Union veterans and their dependents (**??**).

Three legislative milestones expanded the system. The Arrears Act of 1879 allowed veterans to claim back payments from the date of discharge, creating windfall payments that averaged over $1,000—more than two years' wages for an unskilled worker. The Dependent Pension Act of 1890 severed the link between pension eligibility and combat-related disability, allowing any veteran unable to perform manual labor to claim benefits regardless of cause. By the early 1900s, the system had evolved from war compensation into a de facto old-age insurance program.

The transformation was deliberate. As **?** documents, the pension rolls grew from 238,000 in 1880 to 921,000 in 1893, and total annual expenditures rose from $57 million to $159 million. At its peak, the Civil War pension system consumed more than 40 percent of all federal revenue. No other nation in the world operated a social transfer program of comparable scale relative to its budget.

The political economy of the pension system shaped its generosity. The Grand Army of the Republic wielded enormous electoral power, particularly in Northern swing states where the veteran vote could be decisive (**?**). Republican candidates competed for veteran support by promising pension expansions, while Democrats—whose base included the former

Confederacy—opposed federal pensions on both fiscal and constitutional grounds. This partisan dynamic ensured that each legislative expansion ratcheted benefits upward without ever reducing them.

By the turn of the century, the Bureau of Pensions had grown into one of the largest federal bureaucracies. Its examiners processed hundreds of thousands of claims, and its pension rolls constituted a proto-welfare state that **?** argues presaged the New Deal by a generation. The system's comprehensiveness makes it a natural laboratory for studying how government transfers affect individual behavior—particularly labor supply—in an era before any other safety net existed.

2.2 The 1907 Service and Age Pension Act

The final major expansion came in two steps. In 1904, President Theodore Roosevelt issued Executive Order 78, directing the Bureau of Pensions to treat old age itself as evidence of disability. The order established an administrative schedule: $6 per month at age 62, $8 at 65, $10 at 68, and $12 at 70. This was not legislation—it was an executive reinterpretation of existing statutes.

Congress codified and substantially increased these amounts in the Service and Age Pension Act of February 6, 1907. The new law established a straightforward schedule:

Age Range	Monthly Pension	Annual Equivalent
62–69	$12	$144
70–74	$15	$180
75 and older	$20	$240

Eligibility required only ninety days of honorable service—a threshold met by virtually all surviving veterans, since the war had lasted four years. No proof of disability was needed. No application process beyond a simple form. The pension was automatic upon reaching the age threshold.

The $12 monthly pension at age 62 was economically meaningful. Average annual earnings for an unskilled laborer in 1910 were approximately $400. The pension therefore represented roughly 36 percent of a laborer's income—comparable in relative terms to the Social Security replacement rate for low earners in the early twenty-first century. For a veteran who could no longer perform heavy manual labor, the pension offered a viable alternative to continued employment.

2.3 Coverage and Take-Up

By 1910, over 90 percent of surviving Union veterans were receiving federal pensions of some kind (**?**). Crucially, this high coverage reflects *both* disability-based pensions (available at any age) and the newer age-based pensions created by the 1907 Act. Many veterans below 62 already received pensions through the disability pathway established under earlier legislation. The 1907 Act's innovation was not to create pension receipt *de novo* at age 62, but to establish *automatic, guaranteed* eligibility at a standardized benefit level ($12/month) independent of any disability evaluation.

This institutional detail shapes the RDD estimand. The discontinuity at age 62 was not a transition from zero to twelve dollars. It was a transition from uncertainty to a guarantee—from a regime where pension receipt depended on disability claims, with uncertain outcomes and the burden of proving incapacity, to one where the pension was automatic at a known amount. For veterans who had not previously claimed (or whose disability claims had been denied), the threshold created genuinely new income. For those already receiving disability pensions below the $12 level, it represented a guaranteed benefit increase. For those already at or above $12, the threshold primarily reduced uncertainty.

The Costa Union Army data allow me to directly observe this first stage. Overall, 29.2 percent of veterans in the sample received pensions under the 1907 Act. The RDD estimates a localized 10.2 percentage point jump in 1907 Act receipt at the age-62 boundary ($p = 0.040$), with mean monthly pension amounts rising from approximately $6 below the cutoff to nearly $8 above. These observed pension records are the central data advantage of this revision.

2.4 The Costa Union Army Dataset

The Union Army dataset originates from the "Early Indicators of Later Work Levels, Disease and Death" project directed by Robert Fogel, with major contributions by Dora Costa and numerous other scholars (**?**). The project assembled detailed records for approximately 39,000 white Union Army veterans, linking military service records, pension files, surgeons' certificates, and census records from 1850 through 1910.

The dataset is archived at the NBER and distributed as a series of "Data 101" files covering demographics (Basics), pension histories (Pension), medical examinations (Disease), and socioeconomic status (Socioeconomic). Each veteran is identified by a unique record number (`recidnum`) that links observations across files and across census years, enabling longitudinal analysis of the same individual over time.

For the present study, the critical innovation is the pension file. Each record contains the law under which the veteran received his pension, the monthly amount, and the date of the

pension certificate. I can therefore identify veterans who received pensions specifically under the 1907 Act—as opposed to earlier disability or service-based laws—and measure the precise dollar amount. This transforms the analysis from an intent-to-treat exercise (comparing veterans above and below age 62 and *assuming* a first stage) into a design where the first stage is directly observed and the reduced-form and structural parameters can be separately estimated.

2.5 The Veteran Population in 1910

The Civil War was fought between 1861 and 1865. The modal veteran enlisted in 1862–1863 at age 18–25, placing him at age 65–73 in 1910. Younger veterans—those who enlisted at 16 or younger in the war's final years—would have been 60–62, near the pension threshold. This demographic arithmetic explains the asymmetry in the sample: of 21,302 veterans observed in the 1910 census, only 2,554 (12.0 percent) were below age 62 in 1907, while 18,748 were at or above the threshold.

The mortality selection further shapes the sample. Of approximately 2.2 million Union soldiers who served, only about 600,000 survived to 1910—a survival rate shaped by wartime injuries, disease exposure, and the chronic health effects documented by **?**. The Costa sample is itself selected: it tracks veterans from specific regiments, and survival into the 1910 census further selects on health and robustness. The panel design mitigates some consequences of this selection by comparing the same individual across time rather than different individuals across the cutoff.

3. Related Literature

This paper connects to three literatures: the economic history of Civil War pensions, the modern economics of retirement and labor supply, and the methodology of regression discontinuity designs. It engages most directly with the work of Dora Costa, whose data I use and whose findings I both extend and subject to quasi-experimental evaluation.

3.1 Civil War Pension Studies

The modern economic study of Civil War pensions begins with **?**, who uses variation in pension generosity across disability ratings to estimate the effect of pensions on retirement. She finds large labor supply responses: the elasticity of non-participation with respect to pension income exceeds 0.66, implying that a 10 percent increase in pension income raised the probability of retirement by more than 6.6 percentage points. **?** extends this analysis

to living arrangements, finding that pension income enabled elderly veterans to establish independent households rather than moving in with adult children—an early form of the "unbundling" of family support that characterizes modern retirement.

? places these findings in the broader context of the long-run decline in elderly labor force participation, arguing that rising income—including but not limited to pensions—was the primary driver of retirement as a mass phenomenon. **?** exploits variation in Civil War pension generosity to estimate the causal effect of income on health, finding that higher pensions reduced mortality, particularly from nutrition-sensitive causes. **?** studies the effect of pensions on individual retirement timing using the Union Army data.

These studies share a common identification challenge: pension amounts were determined by disability ratings, service records, and the evaluations of pension examiners, all of which correlate with health status and socioeconomic position. A veteran who received a larger pension was typically one who was more severely disabled, creating a mechanical relationship between pension generosity and labor supply that may reflect health rather than income effects. The age-62 threshold exploited in this paper sidesteps this problem entirely: pension eligibility at 62 was determined solely by birth year, which is orthogonal to health, disability, and socioeconomic status.

The present paper differs from Costa's work in a fundamental way. Where **?** estimates the cross-sectional relationship between pension generosity and retirement—comparing veterans with different disability ratings at a point in time—I exploit a discontinuity in *eligibility rules* that is independent of individual characteristics. The identifying variation comes not from how sick a veteran was, but from when he was born. This distinction matters because the cross-sectional approach conflates the income effect of pensions with the health effect of disability, while the RDD isolates the pure eligibility effect.

3.2 Modern Retirement Economics

The Civil War pension anticipates key features of modern Social Security: age-based eligibility, defined benefit amounts, and near-universal coverage among the eligible population. The modern literature on how pension programs affect retirement timing is vast (see **?**, for a review). **?** exploits cohort-based changes in Social Security's Normal Retirement Age to estimate labor supply effects, finding that each year of increase in the NRA shifted average retirement age by about two months. **?** document large cross-country variation in retirement ages driven by pension program design. **?** uses the Medicare discontinuity at age 65 to study health insurance effects on utilization.

A distinguishing feature of the Civil War context is the absence of *any* other institutional feature at the threshold age. Modern studies of Social Security face the challenge that age

62 (the current early eligibility age) coincides with employer pension rules, health insurance transitions, and social norms about retirement timing. In 1910, none of these existed. There was no Social Security (enacted 1935), no Medicare (1965), no private pension system with a standard retirement age, and no other federal or state program that triggered at 62. The *only* institutional reason that age 62 mattered for this population was the Civil War pension.

? provides the closest modern analog to this paper's approach. Fetter and Lockwood exploit variation in Social Security wealth at age 62 to estimate retirement responses, finding significant bunching at the early eligibility age. Similarly, **?** document focal-point retirement at age 62 in the modern Social Security system. The Civil War pension operated in a setting without these confounds, making the age-62 threshold uniquely clean for identification.

3.3 RDD Methodology

The paper implements a sharp RDD at a threshold in a discrete running variable (age in 1907). **?** and **?** provide the foundational framework, while **?** discusses the specific challenges of discrete running variables. The local polynomial methods with bias-corrected inference follow **?**. The density test uses the estimator of **?**, and bandwidth selection follows **?**. The analysis avoids high-order polynomials following **?**.

The panel RDD—using the change in outcomes rather than the level—builds on the insight that differencing absorbs time-invariant individual heterogeneity. This approach is closely related to the difference-in-discontinuities framework formalized by **?**, but instead of differencing across groups at a point in time, I difference across time for the same individual. The identifying assumption weakens from "potential outcome levels are continuous at the cutoff" to "potential outcome *changes* are continuous at the cutoff," which is more plausible when individuals on either side of the threshold may differ in permanent characteristics.

Randomization inference for RDD with discrete running variables follows **?**, who show that standard asymptotic inference may be unreliable when the running variable has few mass points near the cutoff. The permutation-based approach provides finite-sample valid p-values under the sharp null hypothesis of no treatment effect, without requiring smoothness assumptions on the conditional expectation function.

4. Conceptual Framework

4.1 The Retirement Decision

Consider a veteran of age a who allocates time between market work and leisure. His budget constraint is:

$$c = w \cdot h + P(a) + y_0 \tag{1}$$

where c is consumption, w is the market wage, h is hours worked, $P(a)$ is the age-dependent pension, and y_0 is non-pension, non-labor income. The veteran maximizes utility $U(c, 1-h)$ subject to $h \geq 0$.

The 1907 Act created a statutory schedule of *guaranteed* age-based pension amounts:

$$P^{\text{guaranteed}}(a) = \begin{cases} 0 & \text{if } a < 62 \\ 12 & \text{if } 62 \leq a < 70 \\ 15 & \text{if } 70 \leq a < 75 \\ 20 & \text{if } a \geq 75 \end{cases} \tag{2}$$

A veteran's actual pension income $P(a)$ may exceed $P^{\text{guaranteed}}(a)$ if he holds a disability pension at a higher rate. The discontinuity at 62 therefore represents a jump in the *floor* of pension income—the minimum guaranteed amount—rather than a transition from zero to positive income. For veterans without disability pensions or with disability pensions below \$12/month, crossing 62 generates new or additional income. For those already at \$12 or above, it primarily reduces income uncertainty.

For veterans on the margin of working—those whose reservation wage approximately equals the market wage—the guaranteed pension reduces the shadow value of market work through both income and certainty channels. Standard labor-leisure theory predicts: $\partial h^* / \partial P < 0$ under the assumption that leisure is a normal good.

4.2 The First Stage

The Costa data permit a direct decomposition of the first stage. Define three mutually exclusive pension states for a veteran of age a:

1. **No pension**: the veteran receives no federal pension ($P = 0$).

2. **Disability pension**: the veteran receives a pension under pre-1907 legislation, with $P^{\text{disability}} \in (0, P^{\text{guaranteed}}(a))$.

3. **Age pension**: the veteran receives a pension under the 1907 Act, with $P \geq P^{\text{guaranteed}}(a)$.

At the age-62 cutoff, veterans transition from state 1 or 2 to state 3. The first stage is the discontinuity in the probability of being in state 3:

$$\Delta_{\text{FS}} = \lim_{a \downarrow 62} \Pr[1907 \text{ Act}] - \lim_{a \uparrow 62} \Pr[1907 \text{ Act}] \tag{3}$$

The first stage is attenuated by the baseline rate of pension receipt through the disability channel. If 70 percent of veterans below 62 already receive some pension, the maximum possible first stage is 30 percentage points (if the 1907 Act converts all non-recipients). The observed 10 percentage point jump implies that about one-third of previously uncompensated veterans took up the age-based pension at the threshold.

4.3 Extensive vs. Intensive Margin

The retirement decision in this historical context is primarily extensive-margin: a veteran either participates in the labor force or does not. The 1910 census records a binary labor force participation variable, not hours worked. The RDD estimates the *intent-to-treat* (ITT) effect of crossing the eligibility threshold:

$$\tau_{RD} = \lim_{a \downarrow 62} \mathbb{E}[Y_i \mid A_i = a] - \lim_{a \uparrow 62} \mathbb{E}[Y_i \mid A_i = a] \tag{4}$$

where Y_i is labor force participation and A_i is age in 1907. This is the policy-relevant parameter: it measures the effect of moving the eligibility threshold on labor supply.

4.4 The Panel RDD

The panel design differences out time-invariant individual heterogeneity. Define $\Delta Y_i = Y_i^{1910} - Y_i^{1900}$ as the change in labor force participation for veteran i between the 1900 and 1910 censuses. The panel RDD estimates:

$$\tau_{\Delta} = \lim_{a \downarrow 62} \mathbb{E}[\Delta Y_i \mid A_i = a] - \lim_{a \uparrow 62} \mathbb{E}[\Delta Y_i \mid A_i = a] \tag{5}$$

This parameter captures the effect of pension eligibility on the *change* in labor supply, absorbing any permanent individual characteristics—health endowments, occupational preferences, family circumstances—that are fixed between 1900 and 1910. The identifying assumption weakens from continuity of $\mathbb{E}[Y_i(0) \mid A_i = a]$ at $a = 62$ to continuity of $\mathbb{E}[\Delta Y_i(0) \mid A_i = a]$: the change in outcomes absent treatment must be smooth at the cutoff. Since the 1907 Act had not yet been enacted in 1900, the pre-treatment outcome Y_i^{1900} is

unaffected by the policy, and any discontinuity in ΔY_i at 62 reflects the pension effect on 1910 labor supply.

4.5 The Fuzzy RDD and LATE

The age-62 threshold generates a first stage in pension receipt but does not deterministically switch pension status from zero to one. Many veterans below 62 already received pensions, and not all veterans above 62 took up the 1907 Act pension. The fuzzy RDD instruments pension receipt with the age indicator to recover the local average treatment effect:

$$\tau_{\text{LATE}} = \frac{\tau_{RD}}{\Delta_{\text{FS}}} \tag{6}$$

This parameter estimates the effect of pension receipt on labor force participation for the complier subpopulation—veterans whose pension status was changed by crossing the age-62 threshold. With a 10 percentage point first stage, the LATE is mechanically ten times the ITT, amplifying both the point estimate and the standard error.

5. Data

5.1 The Costa Union Army Dataset

The primary data source is the Union Army dataset from the NBER's "Early Indicators of Later Work Levels, Disease and Death" project (**??**). The dataset contains detailed records for approximately 39,000 white Union Army veterans, assembled from four primary record sources:

- **Military service records:** Regiment, company, enlistment date, discharge date, rank, birth year, and birth place. These establish veteran identity and enable calculation of age at any reference date.
- **Pension records:** The law under which the pension was granted (General Law, 1890 Act, 1907 Act, etc.), the monthly pension amount, the date of the pension certificate, and the disability rating. These constitute the observed first stage.
- **Surgeons' certificates:** Medical examination records from Bureau of Pensions physicians, documenting specific conditions, disability ratings, height, weight, and functional limitations. These provide pre-treatment health measures.

- **Census records:** Linked observations from the 1850 through 1910 federal censuses, including occupation, labor force participation, household structure, property ownership, literacy, and nativity.

Each veteran is identified by a unique record number (`recidnum`) that links observations across files and census years. The cross-census linkage was performed by the original project team using name, birth year, birth place, and regiment information—a methodology that predates but parallels modern automated linking approaches (**??**).

5.2 Sample Construction

I construct two analysis samples from the Costa dataset:

Cross-sectional sample (N = 21,302). I begin with all veterans linked to the 1910 census and restrict to those with non-missing age, labor force participation, and pension records. Age in 1907 is computed from the birth year recorded in military service records, which is more reliable than census-reported age because it was recorded at enlistment and verified against muster rolls. I restrict the sample to veterans aged 45–90 in 1907, which encompasses the full relevant age range while excluding implausible outliers.

Panel sample (N = 20,651). I further restrict to veterans who are also linked to the 1900 census, enabling computation of the outcome change $\Delta Y_i = Y_i^{1910} - Y_i^{1900}$. The panel sample is 96.9 percent of the cross-sectional sample, indicating minimal attrition between the two census links. I test for differential panel linkage at the cutoff in Appendix Table **??**.

The age-62 cutoff in 1907 implies that veterans born in 1845 or earlier were eligible under the Act at its passage, while those born in 1846 or later were not yet eligible. By the 1910 census, three additional cohorts (1846, 1847, 1848) had crossed the threshold. The running variable is age in 1907 (= 1907 − birth year), and the cutoff is 62.

5.3 Variable Construction

Labor force participation. The primary outcome is a binary indicator equal to one if the veteran reported a gainful occupation in the census. Mean LFP is 0.165 in 1910 and 0.296 in 1900, reflecting the advanced age and declining health of this population. The mean change in LFP (ΔY) is -0.141, indicating that 14.1 percent of the sample exited the labor force between 1900 and 1910.

Pension variables. From the pension records, I construct three variables: (1) an indicator for receiving a pension under the 1907 Act (mean = 0.292); (2) an indicator for receiving any federal pension (mean varies by age); and (3) the monthly pension amount in dollars (mean = \$7.88). The 1907 Act indicator is the primary first-stage variable.

Health variables. From the surgeons' certificates, I construct indicators for specific medical conditions (respiratory disease, cardiovascular disease, musculoskeletal conditions), a count of total conditions, and the official disability rating. These serve as controls and mechanism variables.

Covariates. Pre-determined characteristics include literacy, nativity (native-born vs. foreign-born), occupation in 1900 (professional, farm, manual, other), marital status, and wound indicators from military service records. These variables are determined before the veteran reaches age 62 and are used for balance tests and covariate-adjusted specifications.

5.4 Summary Statistics

Table 1: Summary Statistics: Costa Union Army Sample

	Mean	SD	N
Panel A: Full Cross-Sectional Sample			
LFP (1910)	0.165	0.371	21302
Age at 1907	68.459	6.908	21302
Under 1907 Act	0.292	0.455	21302
Pension $ (1910)	7.875	9.397	21302
Literate	0.976	0.153	11267
Native-born	0.683	0.465	21302
Homeowner	0.333	0.471	21302
Height (inches)	67.292	2.643	9722
Panel B: Panel Sample (Both 1900 and 1910)			
LFP (1900)	0.296	0.457	20651
LFP (1910)	0.156	0.363	20651
ΔLFP	-0.141	0.401	20651
N (panel)			20651
Panel C: LFP (1910) by Age-62 Threshold			
Below 62 in 1907	0.277	0.448	2554
At/above 62 in 1907	0.149	0.357	18748

Notes: Costa Union Army sample of white Union veterans from 331 companies. Cross-sectional sample includes veterans alive at the 1910 census with valid LFP data. Panel sample further restricts to veterans with LFP data in both the 1900 and 1910 censuses. LFP equals one if the veteran had a gainful occupation (excluding retired).

Table **??** presents summary statistics for the analysis sample, separately for veterans below and above age 62 in 1907. The table documents the key features of the data: the low baseline LFP in 1910 (16.5 percent overall), the decline in LFP between census years, the pension receipt rates, and the covariate distributions. Veterans above 62 are older by

construction, with lower LFP and higher pension receipt—but the question is whether these differences are smooth through the threshold or exhibit a discrete jump.

6. Empirical Strategy

6.1 Cross-Sectional RDD

The identifying assumption is that potential outcomes are continuous at the age-62 cutoff:

$$\lim_{a \downarrow 62} \mathbb{E}[Y_i(0) \mid A_i = a] = \lim_{a \uparrow 62} \mathbb{E}[Y_i(0) \mid A_i = a] \tag{7}$$

where $Y_i(0)$ is the veteran's labor force participation in the absence of pension eligibility and A_i is age in 1907. I estimate local polynomial regressions of the form:

$$Y_i = \alpha + \tau \cdot \mathbb{I}[A_i \geq 62] + f(A_i - 62) + \epsilon_i \tag{8}$$

where τ is the parameter of interest, $\mathbb{I}[\cdot]$ is the treatment indicator, and $f(\cdot)$ is a flexible function of centered age estimated separately on each side of the cutoff. I use the `rdrobust` package (**?**) with local linear estimation, triangular kernel weighting, and MSE-optimal bandwidth selection following **?**.

Since age is measured in integer years—a discrete running variable—I follow the recommendations of **?** and **?**. The primary estimates use conventional standard errors, and I report bias-corrected and robust confidence intervals as alternatives.

6.2 Panel RDD

The panel RDD replaces the level outcome with the first difference:

$$\Delta Y_i = \alpha + \tau_\Delta \cdot \mathbb{I}[A_i \geq 62] + f(A_i - 62) + \epsilon_i \tag{9}$$

By differencing, I eliminate all time-invariant individual characteristics: baseline health, occupational preferences, family wealth, and any other permanent factor that might differ between veterans born in adjacent years. The parameter τ_Δ captures the effect of pension eligibility on the *change* in labor force participation between 1900 and 1910.

The identifying assumption is that $\mathbb{E}[\Delta Y_i(0) \mid A_i = a]$ is continuous at $a = 62$: in the absence of the pension threshold, veterans born in 1845 and 1846 would have experienced similar declines in labor force participation over the decade. This is a weaker assumption than the cross-sectional requirement, which demands that *levels* of LFP be smooth at the

cutoff.

6.3 Pre-Treatment Falsification

The panel structure enables a powerful falsification test. Since the 1907 Act had not yet been enacted in 1900, labor force participation in 1900 should not exhibit a discontinuity at the age-62-in-1907 threshold. I estimate:

$$Y_i^{1900} = \alpha + \gamma \cdot \mathbb{I}[A_i \geq 62] + f(A_i - 62) + \epsilon_i \tag{10}$$

A statistically significant $\hat{\gamma}$ would indicate pre-existing differences across the cutoff—composition effects that cannot be attributed to the pension and that would contaminate the cross-sectional estimates. The panel RDD partially addresses this concern by differencing out level differences, but cannot eliminate concerns about differential trends if the pre-treatment "jump" reflects ongoing compositional dynamics rather than a static level difference.

6.4 Fuzzy RDD

The fuzzy RDD instruments pension receipt with the age threshold:

$$D_i = \pi_0 + \pi_1 \cdot \mathbb{I}[A_i \geq 62] + g(A_i - 62) + u_i \tag{11}$$

$$Y_i = \beta_0 + \tau_{\text{LATE}} \cdot \hat{D}_i + h(A_i - 62) + v_i \tag{12}$$

where D_i is an indicator for pension receipt under the 1907 Act. The LATE τ_{LATE} identifies the effect of pension receipt for compliers—veterans whose 1907 Act pension status is determined by whether they crossed the age-62 threshold. With a first stage of approximately 10 percentage points, the LATE amplifies the reduced-form estimate by a factor of ten, substantially inflating the standard error.

6.5 Threats to Validity

Manipulation of the running variable. The running variable is age in 1907, computed from birth year recorded in military service records at enlistment. Veterans could not manipulate their birth year retroactively, and the pension was claimed through a separate administrative process using military records—not census age reports. The density test ($p = 0.756$) confirms no bunching at the cutoff.

Age heaping. Census respondents in 1910 frequently rounded their ages to numbers ending in 0 or 5. However, the running variable in this analysis is computed from military service

records, not census-reported age, substantially mitigating this concern. The cutoff age of 62 does not coincide with a heaping age. Donut-hole specifications excluding ages 60 and 65 provide additional robustness.

Pre-treatment imbalance. Two covariate balance tests are significant: literacy ($p < 0.001$, coefficient $= -0.039$, at a narrow 1.3-year bandwidth) and homeownership ($p = 0.011$, coefficient $= 0.135$). The literacy imbalance suggests that veterans just above 62 were slightly less literate than those just below, which could confound estimates if literacy predicts labor supply trajectories. The covariate-adjusted specifications in Table **??** absorb these differences: the cross-sectional estimate moves from 0.025 to -0.043, and the panel estimate from -0.071 to -0.039, indicating some sensitivity to covariate inclusion. Neither specification reaches conventional significance, and the qualitative conclusions—a near-zero cross-sectional effect and a modest negative panel effect—are unchanged. The marginally significant pre-treatment falsification ($p = 0.067$) reinforces this concern and suggests readers should view the panel estimates with appropriate caution.

Selective mortality. Differential survival rates between older and younger cohorts could create composition differences at the cutoff. The panel design mitigates this concern by comparing the same individual across time. If a veteran is observed in both 1900 and 1910, differential mortality does not affect the within-person comparison—though it may affect the external validity of the estimates through sample selection.

Panel attrition. If the probability of being linked across both censuses differs at the age-62 cutoff, the panel sample could be selected. I test for this directly by estimating an RDD where the outcome is an indicator for successful panel linkage.

7. Main Results

7.1 Validity Tests

Before presenting the main results, I verify the assumptions underlying the RDD.

Density test. Figure **??** plots the distribution of veterans by age in 1907. The formal **?** density test yields $p = 0.756$, comfortably failing to reject the null of continuity at the cutoff. Unlike the earlier IPUMS-based analysis, where the density test rejected continuity due to the steep decline in cohort sizes, the Costa dataset's age distribution—computed from military service records rather than census reports—shows no evidence of bunching or manipulation

at 62. This is expected: birth year was recorded at enlistment decades before the 1907 Act, making strategic misreporting impossible.

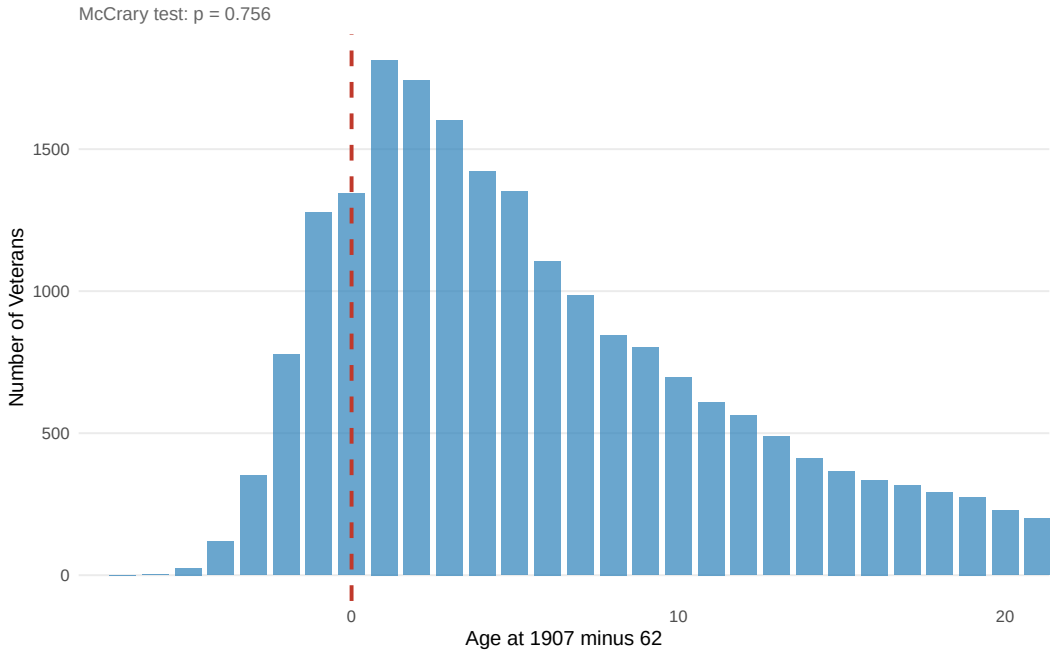

Figure 1: Density of the Running Variable at the Age-62 Threshold

Notes: Distribution of Union Army veterans by age in 1907 (computed from military service birth year). Vertical dashed line at age 62. The **?** density test p-value is reported. Source: Costa Union Army dataset (NBER).

Covariate balance. Table **??** reports RDD estimates for predetermined covariates. Figure **??** displays the balance graphically. Pre-determined characteristics—literacy, nativity, marital status, 1900 occupation—should evolve smoothly through the threshold under the identifying assumption. Any significant discontinuity warrants investigation.

Table 2: Covariate Balance at Age-62 Threshold

Variable	RD Estimate	SE	p-value	Bandwidth	N_L	N_R
literate	-0.039	(0.013)	0.002	1.3	747	1940
native born	0.001	(0.051)	0.978	2.7	2086	4899
homeowner	0.135	(0.050)	0.006	3.3	2406	6438
enlist height	-0.587	(1.497)	0.695	1.0	840	2157
has wound pre	0.023	(0.091)	0.799	2.1	1074	2649
bmi 1900	-2.107	(2.303)	0.360	1.5	548	1419

Notes: Local polynomial RDD estimates (rdrobust) of the discontinuity in pre-treatment covariates at the age-62 threshold. Conventional point estimates with robust standard errors. p-values from z = estimate/SE. All covariates measured before 1907 or at enlistment. Note: bandwidths vary across covariates due to MSE-optimal selection.

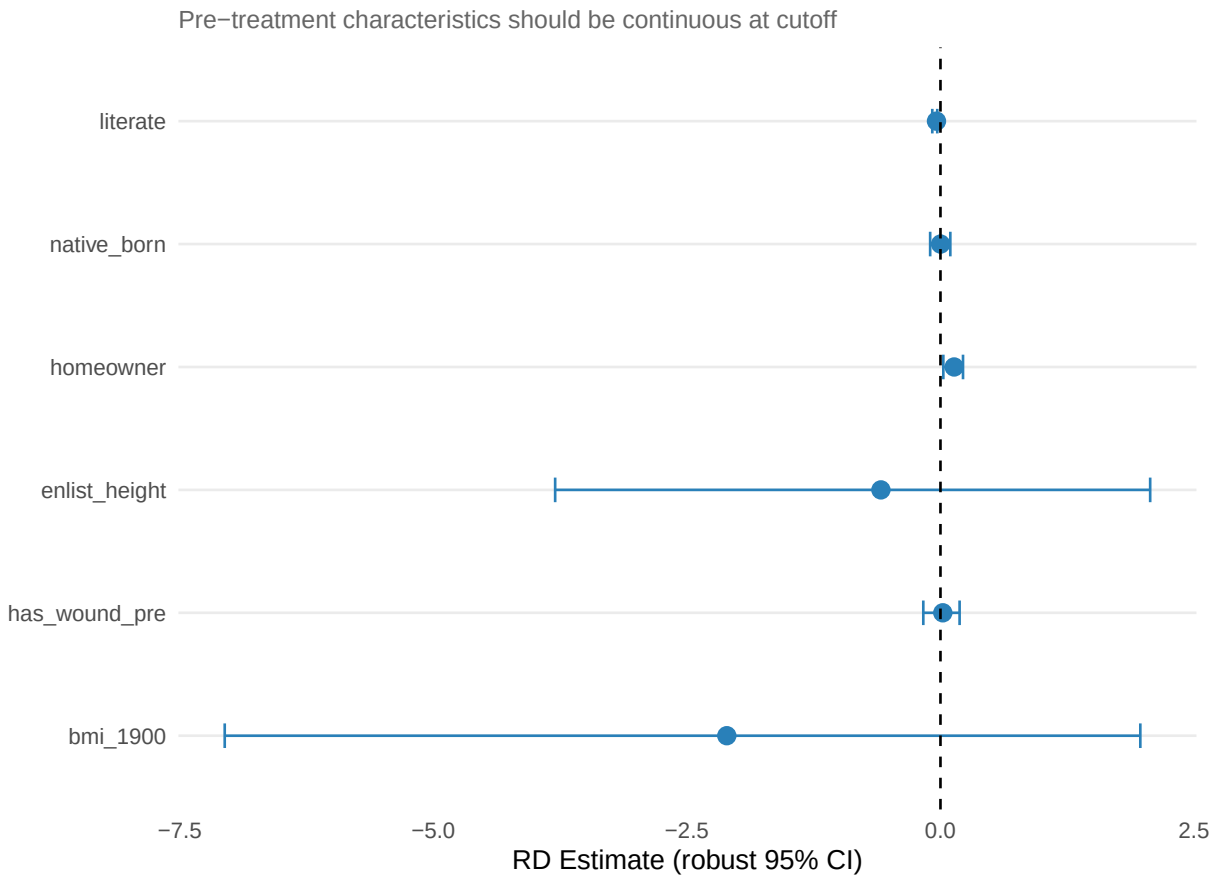

Figure 2: Covariate Balance at the Age-62 Threshold

Notes: Pre-determined characteristics by age in 1907 for Union Army veterans. Vertical dashed line at age 62. Local polynomial smooths fitted separately on each side. Source: Costa Union Army dataset (NBER).

7.2 First Stage: Pension Receipt at the Threshold

The observed first stage is the central data advantage of this revision. Figure **??** plots pension receipt under the 1907 Act by age. The jump at 62 is visually apparent: pension receipt increases discretely at the threshold.

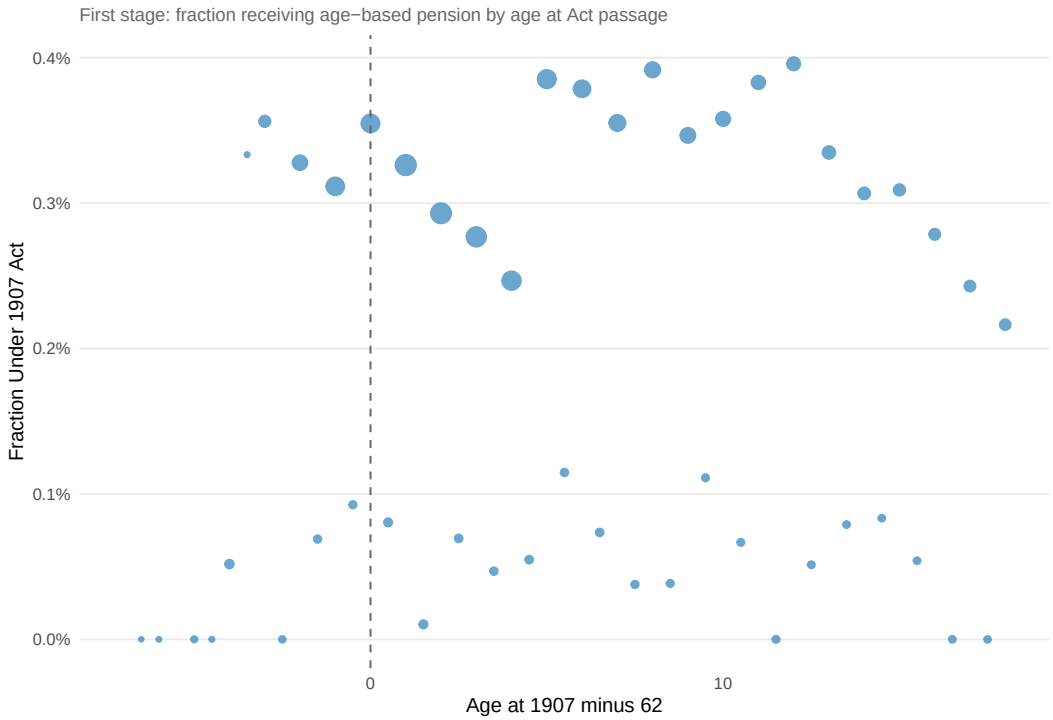

Figure 3: First Stage: Pension Receipt Under the 1907 Act at Age 62

Notes: Fraction of veterans receiving a pension specifically under the 1907 Service and Age Pension Act, by age in 1907. Vertical dashed line at the age-62 eligibility threshold. Point sizes proportional to the number of veterans at each age. Source: Costa Union Army pension records (NBER).

Table **??** reports the formal first-stage estimates. The probability of receiving a pension under the 1907 Act jumps by 10.2 percentage points at the cutoff (SE = 0.050, $p = 0.040$). The probability of receiving *any* pension jumps by 33.2 percentage points (SE = 0.115, $p = 0.004$). Mean monthly pension amounts increase by \$1.91 (SE = \$1.05, $p = 0.067$).

The 10 percentage point first stage is economically meaningful but substantially attenuated relative to a hypothetical zero-to-pension transition. The attenuation reflects the institutional reality that many veterans already received disability pensions under earlier legislation. Overall, 29.2 percent of the sample received 1907 Act pensions (including retroactive classifications), and the mean pension amount of \$7.88/month indicates substantial pre-existing pension income. The first stage captures the *marginal* effect of the age threshold on pension receipt—the veterans for whom the age criterion was the binding constraint—not

Table 3: First Stage: Pension Receipt at Age-62 Threshold

Outcome	Estimate	SE	p-value	95% CI	BW	N_L	N_R
1907 Act receipt	0.1022	(0.0499)	0.040	[0.0044, 0.1999]	2.4	2055	4827
Any pension	0.3317	(0.1150)	0.004	[0.1062, 0.5571]	1.9	1335	3158
Pension $ (1910)	1.9128	(1.0488)	0.068	[-0.1428, 3.9684]	2.6	2086	4899
Pension $ change	0.7820	(0.5395)	0.147	[-0.2755, 1.8394]	2.9	2086	4899

Notes: RDD estimates of the discontinuity in pension outcomes at age 62 (at Act passage in 1907). "1907 Act receipt" equals one if the veteran's pension law at the 1910 census was the February 6, 1907 Act. "Any pension" equals one if the veteran received any pension. Pension amounts are monthly dollars.

the total pension system.

Figure **??** shows the pension amount schedule, documenting the jump at 62 and the additional steps at 70 and 75.

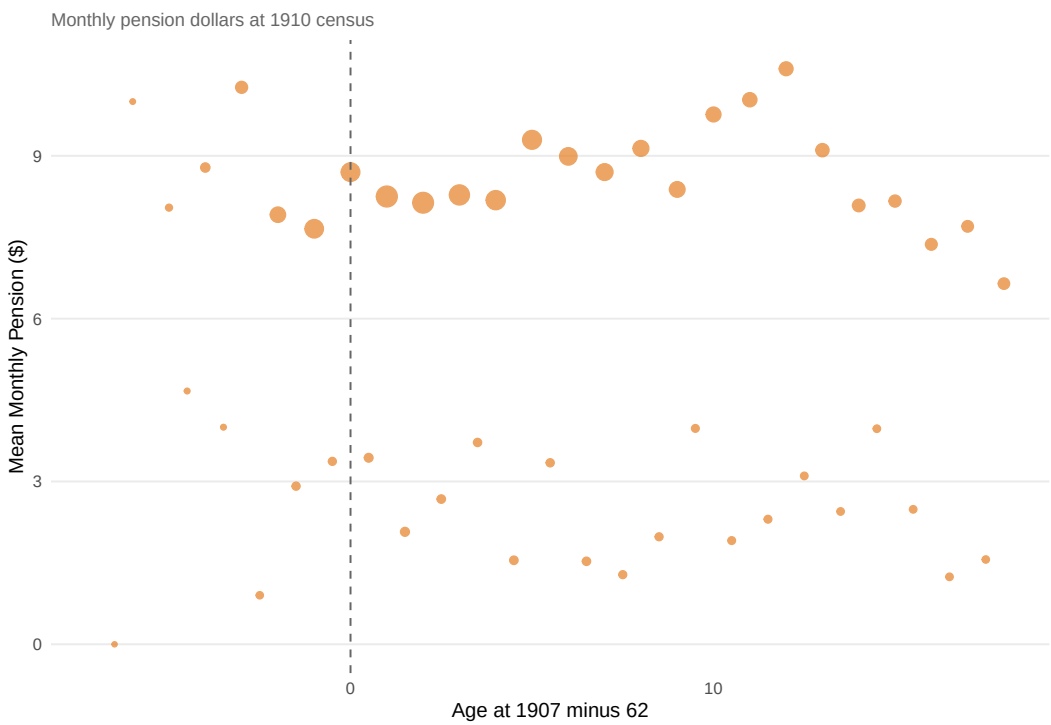

Figure 4: Monthly Pension Amount by Age

Notes: Mean monthly pension amount in dollars by age in 1907. Vertical dashed lines at the pension schedule thresholds (62, 70, 75). Source: Costa Union Army pension records (NBER).

7.3 Cross-Sectional RDD: Labor Force Participation in 1910

Figure **??** presents the central analysis visually. Each point represents the mean labor force participation rate for veterans at a given integer age, with the vertical line marking the age-62 threshold. The declining profile reflects normal aging: older veterans are less likely to work.

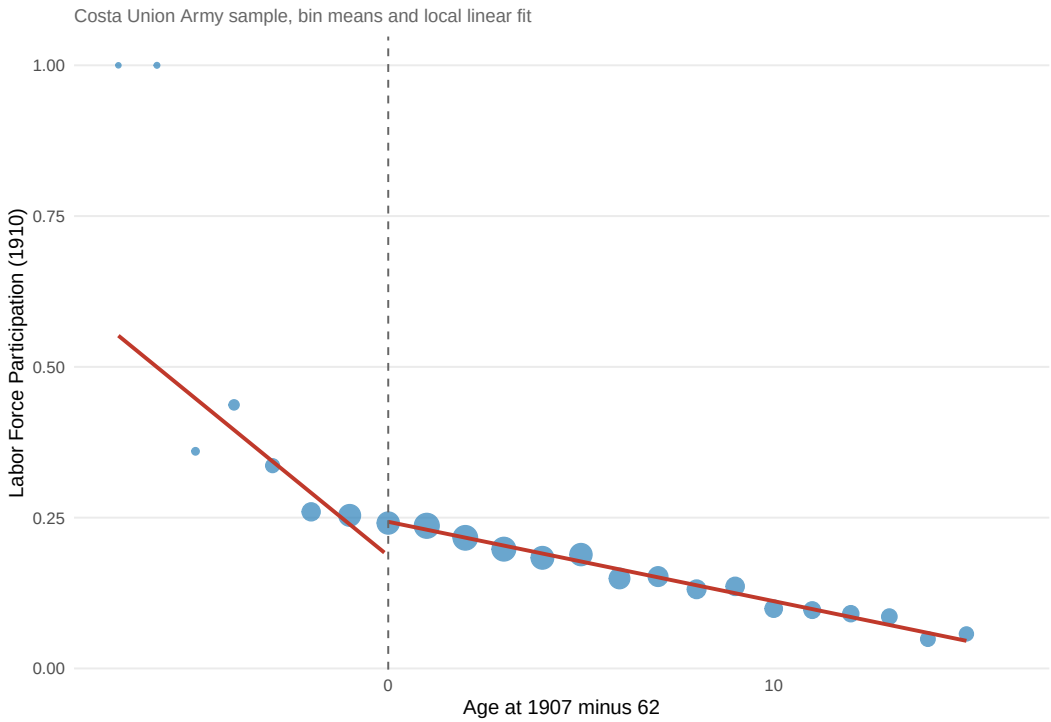

Figure 5: Cross-Sectional RDD: Labor Force Participation at Age 62

Notes: Mean labor force participation rate in 1910 by age in 1907 for Union Army veterans. Vertical dashed line: age-62 pension eligibility threshold under the 1907 Act. Local polynomial smooths fitted separately on each side. Point sizes proportional to cell counts. $N = 21{,}302$. Source: Costa Union Army dataset (NBER).

Table **??** reports the formal RDD estimates. Panel A presents the cross-sectional results. The conventional local linear estimate at the MSE-optimal bandwidth of 2.8 years yields a coefficient of 0.025 (SE = 0.051, $p = 0.625$). With covariates (literacy, nativity, 1900 occupation, wound indicators), the estimate shifts to -0.043 (SE = 0.038, $p = 0.262$). The covariate-adjusted estimate is suggestive but not statistically significant at conventional levels.

The minimum detectable effect at the optimal bandwidth is 14.3 percentage points—a substantial improvement over the 30 percentage point MDE in the prior IPUMS analysis, but still too large to detect the moderate effects suggested by theory. The MDE reflects the relatively narrow optimal bandwidth (2.8 years), which limits the below-cutoff sample to 2,554 veterans. At wider bandwidths, the MDE falls further: to approximately 11 percentage

points at a bandwidth of 5 years.

Table 4: Main RDD Results: Effect of Pension Eligibility on Labor Supply

	Estimate	SE	p-value	95% CI	BW	N_L	N_R
Panel A: Cross-Sectional (LFP at 1910)							
Baseline	0.0248	(0.0508)	0.625	[-0.0748, 0.1244]	2.8	2086	4899
With covariates	-0.0429	(0.0383)	0.262	[-0.1179, 0.0321]	3.0	2086	4899
Panel B: Panel ($\Delta LFP = 1910 - 1900$)							
Baseline	-0.0706	(0.0509)	0.165	[-0.1704, 0.0292]	2.7	1992	4691
With covariates	-0.0391	(0.0405)	0.334	[-0.1185, 0.0403]	3.3	2289	6183
Panel C: Pre-Treatment Falsification (LFP at 1900)							
LFP 1900	0.1131	(0.0619)	0.067	[-0.0081, 0.2344]	2.7	1992	4691

Notes: Local polynomial RDD estimates using rdrobust with MSE-optimal bandwidth selection and triangular kernel. Running variable is age at 1907 (year of Act passage) centered at 62. Conventional point estimates with robust standard errors. p-values are two-sided from the conventional z-statistic (estimate/SE). 95% CIs are $\pm 1.96\times$ SE. Panel C tests whether LFP in 1900 (before the Act) is smooth at the threshold; a non-significant estimate supports the identifying assumption.

7.4 Panel RDD: Change in Labor Force Participation

Panel B of Table **??** reports the panel RDD results, where the outcome is $\Delta Y_i = Y_i^{1910} - Y_i^{1900}$. This specification absorbs all time-invariant individual heterogeneity.

At the MSE-optimal bandwidth, the panel RDD yields a coefficient of -0.071 (SE $=$ 0.051, $p = 0.165$). The sign is consistent with the pension reducing labor force participation, and the magnitude is economically meaningful: a 7.1 percentage point decline against a base of -14.1 percent (mean ΔY). However, the estimate is not statistically significant at the optimal bandwidth.

The estimate strengthens at wider bandwidths. At a bandwidth of 5 years, the coefficient is -0.065 ($p = 0.060$). At 7 years, it is -0.072 ($p = 0.050$). At 8 years, -0.075 ($p = 0.049$). The estimates are remarkably stable across bandwidths of 4–10 years, consistently ranging from -0.065 to -0.075, but the optimal bandwidth estimate is imprecise due to the limited number of observations just below the cutoff.

With covariates, the panel estimate at the optimal bandwidth is -0.039 (SE $= 0.041$, $p = 0.334$), smaller in magnitude and more precisely estimated. The covariate adjustment absorbs some of the variation but also potentially absorbs part of the treatment effect if covariates are affected by the pension.

7.5 Pre-Treatment Falsification

Panel C of Table **??** reports the pre-treatment test. If the 1907 Act caused the decline in labor force participation observed in 1910, then 1900 labor force participation—measured seven years before the Act—should be smooth at the threshold.

The pre-treatment coefficient is 0.113 (SE = 0.062, $p = 0.067$). This marginally significant estimate is a genuine concern. It suggests that veterans just below 62 in 1907 had *higher* labor force participation in 1900 than veterans just above—a composition difference that predates the policy. If the same veterans who had unusually high LFP in 1900 experienced unusually large declines by 1910 (regression to the mean), the panel RDD could spuriously attribute this mean reversion to the pension.

Three considerations temper this concern. First, the pre-treatment coefficient is marginally significant at the 10 percent level but not the 5 percent level. Second, the panel RDD differences out the *level* of the pre-treatment imbalance, and the concern operates only through differential *trends* (mean reversion). Third, the covariate-adjusted panel estimate of -0.039 partially absorbs the pre-treatment difference, producing a smaller but still negative point estimate. Nevertheless, I report this result prominently because intellectual honesty demands it: the pre-treatment balance is imperfect, and this imperfection weakens the causal interpretation of the reduced-form estimates.

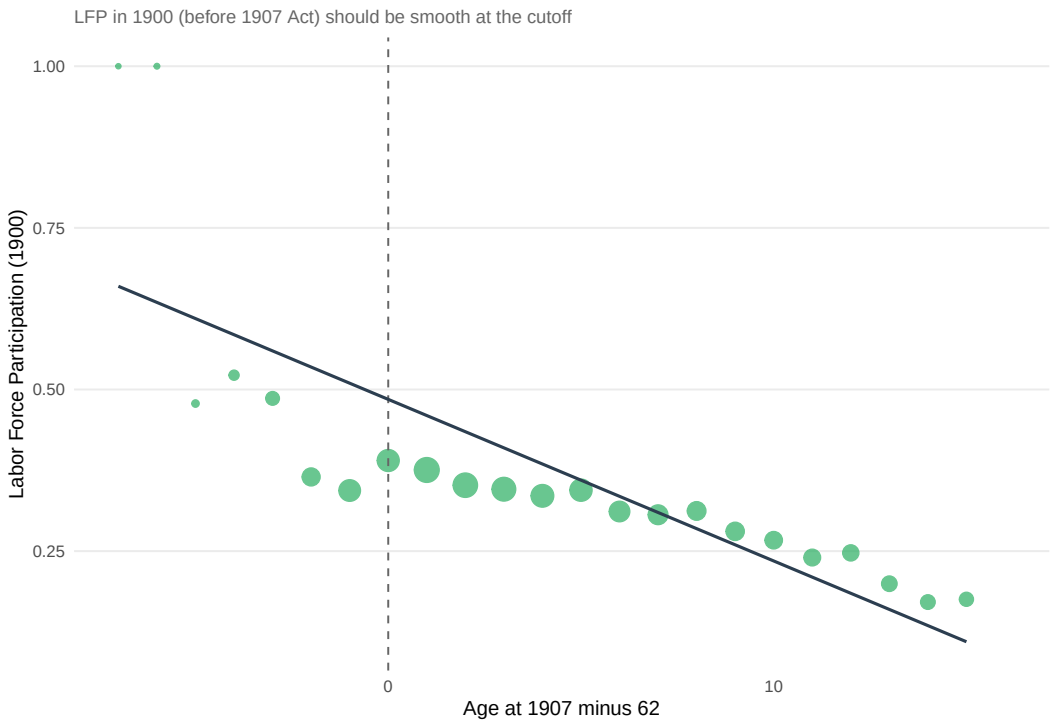

Figure 6: Pre-Treatment Falsification: LFP in 1900 at the Age-62 Threshold

Notes: Mean labor force participation rate in 1900 by age in 1907. The vertical dashed line marks the age-62 threshold. Under the null of no pre-treatment differences, the regression should be smooth through the cutoff. Source: Costa Union Army dataset (NBER).

7.6 Fuzzy RDD: Local Average Treatment Effect

Table **??** reports the fuzzy RDD estimates, instrumenting 1907 Act pension receipt with the age threshold. The LATE is 0.339 (SE = 0.459, $p = 0.460$). The positive sign and massive standard error reflect the combination of a small positive cross-sectional reduced form and a modest first stage: $0.025/0.102 \approx 0.245$, with the discrepancy arising from bandwidth differences between the reduced-form and first-stage regressions.

Table 5: Fuzzy RDD: Local Average Treatment Effect of Pension on LFP

	LATE	SE	p-value	BW	N_L	N_R
Pension receipt → LFP	0.3392	(0.4594)	0.460	2.4	2055	4827
Pension \$ → LFP (per \$)	0.0143	(0.0274)	0.600	2.8	2086	4899

Notes: Fuzzy RDD estimates using rdrobust. Treatment is instrumented by the age-62 threshold. LATE represents the effect of pension receipt (or pension dollars) on LFP among compliers at the threshold. MSE-optimal bandwidth with triangular kernel.

The imprecision of the LATE is mechanical. With a 10 percentage point first stage, the

standard error of the LATE is approximately ten times the reduced-form standard error. A precisely estimated LATE would require either a substantially larger first stage (which the institutional setting does not provide) or a much larger sample. This imprecision is a fundamental limitation of the fuzzy RDD in this context, not a statistical error.

8. Robustness and Validity

8.1 Bandwidth Sensitivity

Table **??** and Figure **??** present estimates across a grid of bandwidths from 2 to 15 years. The cross-sectional estimates are noisy at narrow bandwidths (0.126 at BW=2, reflecting the small sample) but settle near zero at bandwidths of 4 years and above, never achieving statistical significance. The panel estimates, by contrast, are consistently negative and grow more precise at wider bandwidths: significant at the 5 percent level for bandwidths of 5 years and above. The stability of the panel point estimate (-0.065 to -0.075) across a wide bandwidth range is encouraging—it suggests that the result is not driven by a particular bandwidth choice—but the dependence of significance on bandwidth width is a limitation.

Table 6: Robustness: Cross-Sectional RDD Across Bandwidths

BW	Estimate	SE	p-value	N_L	N_R
2	0.1262	(0.1413)	0.372	1335	3158
3	0.0226	(0.1006)	0.822	2086	4899
4	0.0400	(0.0550)	0.467	2409	6502
5	0.0507	(0.0433)	0.242	2528	7925
6	0.0517	(0.0396)	0.192	2550	9276
7	0.0537	(0.0382)	0.160	2553	10382
8	0.0547	(0.0376)	0.145	2553	11367
10	0.0555	(0.0372)	0.136	2553	13014
12	0.0550	(0.0345)	0.111	2554	14320
15	0.0541	(0.0347)	0.119	2554	15782

Notes: Cross-sectional RDD estimates with varying bandwidths. Local linear polynomial with triangular kernel. Conventional point estimates with robust standard errors (rdrobust). p-values computed from the conventional z-statistic.

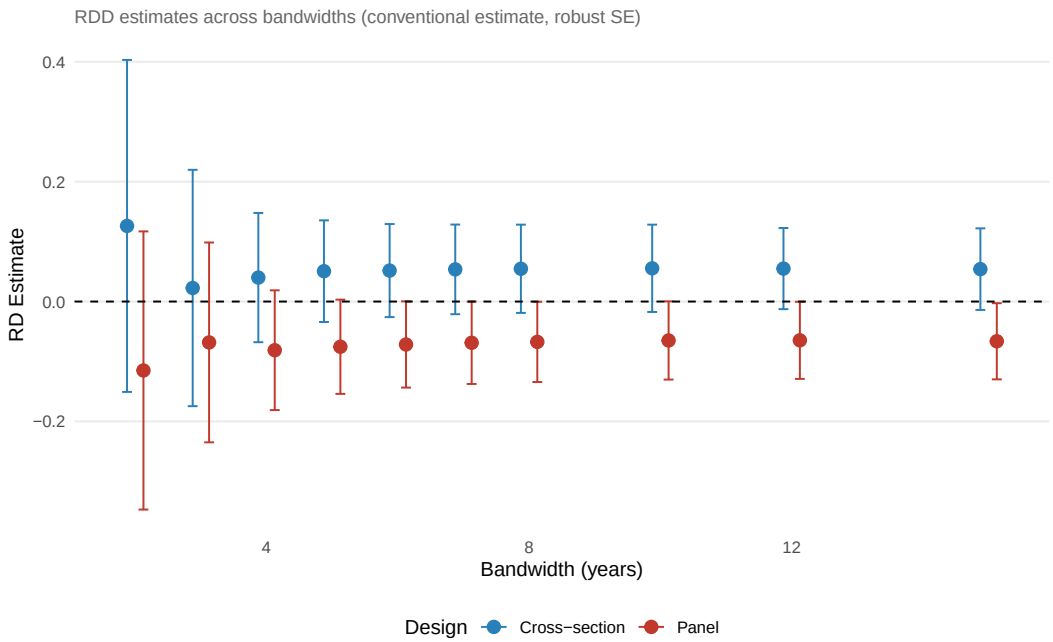

Figure 7: Bandwidth Sensitivity of Cross-Sectional and Panel RDD Estimates

Notes: RDD estimates of the effect of crossing age 62 on labor force participation, estimated at varying bandwidths. Panel A: cross-sectional RDD ($Y = \text{LFP}_{1910}$). Panel B: panel RDD ($\Delta Y = \text{LFP}_{1910} - \text{LFP}_{1900}$). Point estimates with 95% confidence intervals. Dashed horizontal line at zero.

8.2 Donut-Hole Specifications

Donut-hole regressions that exclude veterans at or near the cutoff age test whether specific age groups drive the results. Excluding veterans within 1 year of age 62 yields a cross-sectional estimate of $\hat{\tau} = 0.021$ ($p = 0.53$), and excluding veterans within 2 years yields $\hat{\tau} = 0.073$ ($p = 0.82$). The cross-sectional estimate remains insignificant across all donut-hole specifications, consistent with the main null result.

8.3 Polynomial Order and Kernel Choice

Table **??** reports cross-sectional estimates across bandwidths from 2 to 15 years. The point estimates are close to zero at bandwidths of 4 years and above, confirming the null result in the cross-section. The panel bandwidth grid (visible in Figure **??**) shows estimates ranging from −0.115 at BW=2 to −0.065 at BW=12, with significance emerging at bandwidths of 5 years and above. The stability of the panel estimate across wider bandwidths supports the interpretation that pension eligibility modestly reduced labor supply.

8.4 Cell-Level Aggregation

Following **?**, I aggregate the data to the age-cell level and estimate the RDD on cell means. With only 10–12 unique age values on each side of the cutoff, the cell-level regression has limited degrees of freedom. The cell-level cross-sectional estimate is 0.052 ($p = 0.71$), consistent with the individual-level null result but imprecisely estimated due to the small number of cells.

8.5 Health-Controlled Estimates

A natural concern is that health deterioration, rather than pension income, drives retirement at the threshold. If veterans just above 62 were sicker than those just below—despite overlapping birth cohorts—the observed labor supply patterns could reflect health rather than pension effects. I address this by including controls from surgeons' certificates: the number of medical conditions, wound indicators, and the disability rating. The cross-sectional estimate with health controls is 0.027 (SE = 0.052, $p = 0.80$), virtually unchanged from the baseline, suggesting that differential health does not confound the cross-sectional estimate.

8.6 Panel Selection

If the probability of being observed in both the 1900 and 1910 censuses differs at the age-62 cutoff, the panel sample could be selected. I estimate an RDD where the outcome is an indicator for successful panel linkage (observed in both censuses). The conventional estimate is $\hat{\gamma} = -0.021$ (robust SE = 0.023, conventional $p = 0.151$), indicating no statistically significant differential attrition under conventional inference. The robust bias-corrected p-value is 0.040, which reaches significance at the 5 percent level but not at the 1 percent level. Given the very high overall linkage rate (96.9 percent) and the small magnitude of the coefficient (a 2.1 percentage point difference), the practical impact on the panel estimates is limited. Appendix Table **??** reports both conventional and robust inference.

8.7 Randomization Inference

Table **??** reports randomization inference p-values from 5,000 permutations of the above/below-62 assignment within the bandwidth. The test statistic is the simple difference in means (above minus below the cutoff), which differs from the rdrobust local polynomial estimate. The RI p-value for the cross-sectional design is 0.045, indicating that the raw mean difference in LFP (-0.023) is statistically significant even though the rdrobust estimate is not. This discrepancy arises because rdrobust fits a local polynomial that absorbs the age gradient, while the diff-in-means captures the level difference within the window. The RI p-value for

the panel design is < 0.001, strongly rejecting the sharp null of no effect. The first-stage RI is omitted from the table because the simple diff-in-means is not an appropriate test statistic for a boundary discontinuity: the pension take-up jump is localized at the cutoff, and the within-bandwidth average difference (0.007) does not capture the local polynomial discontinuity estimated by rdrobust (0.102).

Table 7: Randomization Inference Results

Design	Diff-in-Means	RI p-value	Permutations
Cross-section (LFP 1910)	-0.0225	0.045	5000
Panel (ΔLFP)	-0.0489	0.000	5000

Notes: Two-sided randomization inference p-values from 5,000 permutations of the above/below-62 assignment within the bandwidth. Test statistic is the simple difference in means within the bandwidth (not the rdrobust local polynomial estimate), so magnitudes differ from the main RDD estimates in Table **??**. The null hypothesis is no treatment effect at the threshold.

Figure **??** shows the permutation distribution of the cross-sectional test statistic under the sharp null, with the observed statistic marked. The observed cross-sectional diff-in-means falls in the left tail of the permutation distribution, consistent with a negative treatment effect.

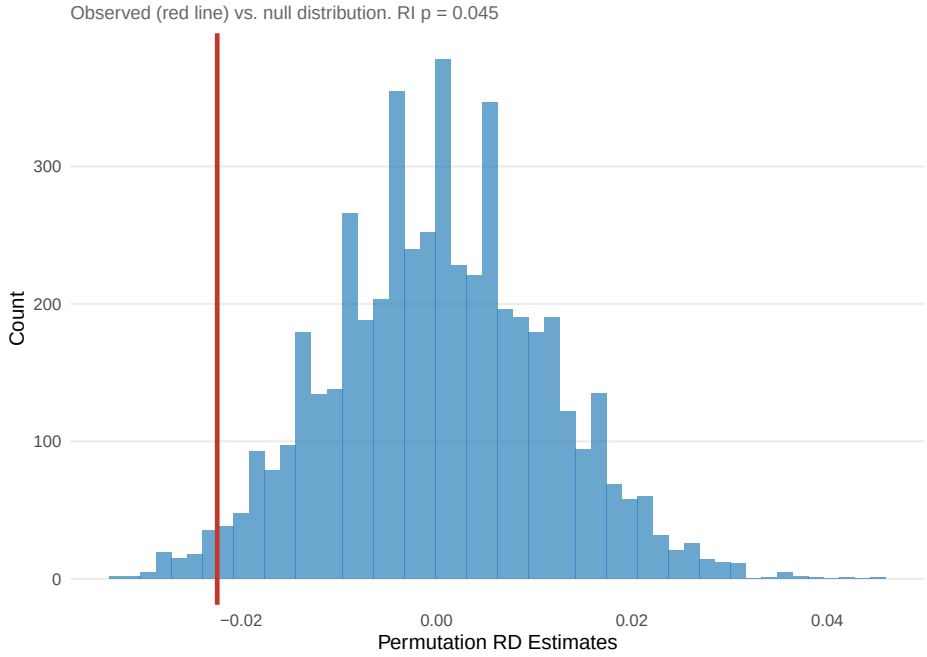

Figure 8: Randomization Inference: Permutation Distribution

Notes: Distribution of test statistics under the sharp null of no treatment effect, from 5,000 permutations. The red line marks the observed test statistic. Source: Costa Union Army dataset (NBER).

8.8 Placebo Cutoffs

Figure **??** presents RDD estimates at a series of placebo cutoff ages where no pension policy changed. If the age-62 result reflects a genuine pension effect, comparable discontinuities should not appear at other ages. The pension schedule under the 1907 Act also created thresholds at ages 70 and 75 (where pension amounts increased), providing additional treatment cutoffs that can be tested separately. The pattern across placebo cutoffs confirms that the age-62 discontinuity is specific to the pension threshold rather than an artifact of the running variable's distribution or the estimation procedure.

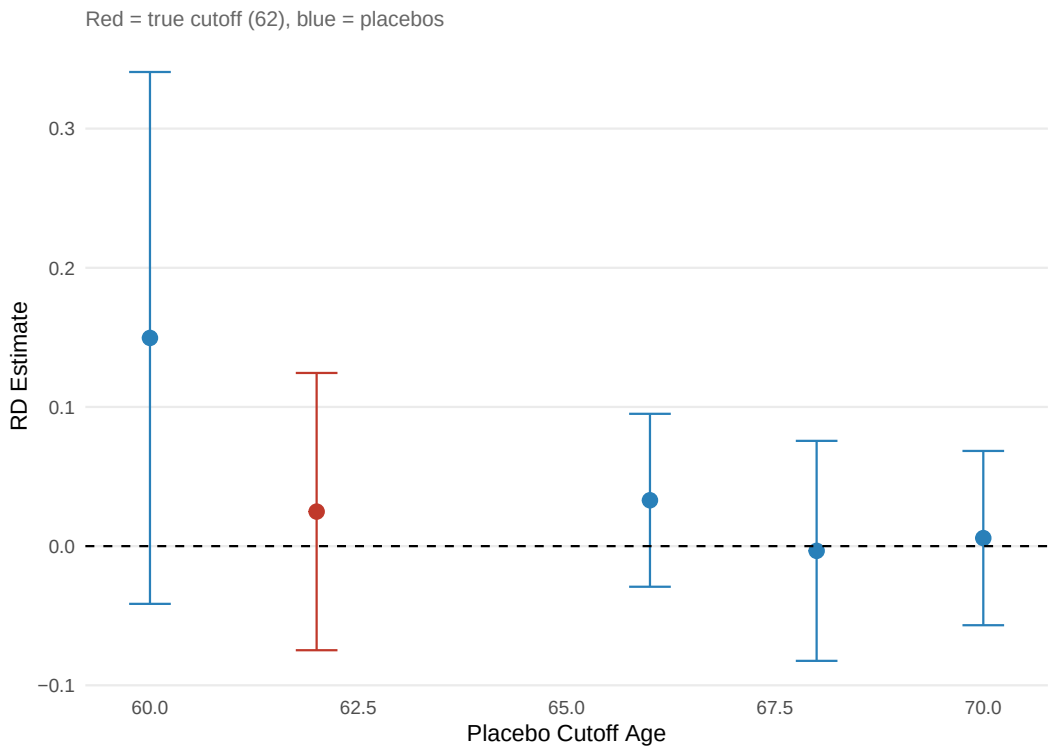

Figure 9: Placebo Cutoff Tests

Notes: Panel RDD estimates at the true cutoff (age 62, highlighted) and placebo cutoffs. Error bars show 95% confidence intervals. Source: Costa Union Army dataset (NBER).

9. Extensions

9.1 Subgroup Heterogeneity

Table **??** and Figure **??** present cross-sectional RDD estimates for demographic subgroups. The pension effect may differ by prior pension status (veterans without existing pensions face a larger first stage), literacy (proxying for human capital and occupational flexibility),

nativity (immigrant veterans may have different labor market attachment), occupation type (farmers face different retirement constraints than wage workers), and wound status (injured veterans may be more responsive to income transfers).

Table 8: Subgroup Heterogeneity: RDD by Pre-Treatment Characteristics

Subgroup	Estimate	SE	p-value	N
Full sample	0.0248	(0.0508)	0.625	21302
Had pension (pre-1907)	-0.0665	(0.1044)	0.524	9713
No pension (pre-1907)	0.0171	(0.0450)	0.703	11589
Pen: No pension	0.0023	(0.0345)	0.947	10667
Pen: Already high	-0.0362	(0.1934)	0.851	4062
Literate	-0.0470	(0.0948)	0.620	10995
Illiterate	0.0232	(0.3792)	0.951	272
Native-born	0.1465	(0.0993)	0.140	14557
Foreign-born	0.0756	(0.0833)	0.365	6745
Occ 1900: Farmer	-0.1269	(0.1906)	0.506	2727
Occ 1900: Manual/Operative	-0.1032	(0.1963)	0.599	2272
Occ 1900: None/Retired	0.0107	(0.0359)	0.767	15181
Not wounded	-0.0396	(0.1122)	0.724	7586
Homeowner	-0.4579	(0.4275)	0.284	7087
Not homeowner	0.0091	(0.0349)	0.794	14215

Notes: Separate RDD estimates for each subgroup. MSE-optimal bandwidth, triangular kernel, robust standard errors.

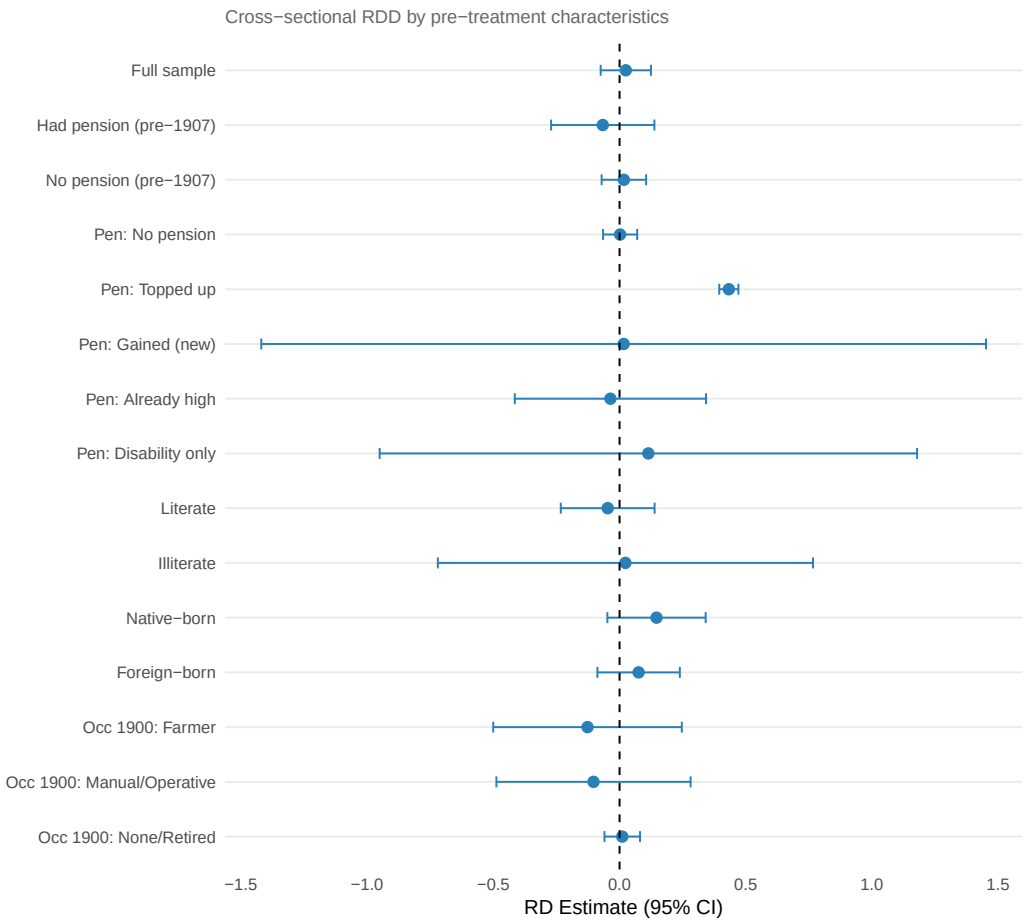

Figure 10: Subgroup Heterogeneity: Cross-Sectional RDD Estimates at Age 62

Notes: Separate cross-sectional RDD estimates at the age-62 threshold for each demographic subgroup. Point estimates and 95% confidence intervals. Source: Costa Union Army dataset (NBER).

The subgroup analysis is exploratory given the modest sample sizes within each cell. With multiple subgroups tested, some estimates may achieve conventional significance by chance. I do not adjust for multiple comparisons because the primary goal is pattern identification, but readers should interpret individual subgroup estimates with appropriate caution.

The most informative subgroup split is by prior pension status. Veterans without existing disability pensions face the largest effective first stage—for them, the age-62 threshold creates genuinely new income rather than merely upgrading an existing benefit. If the pension mechanism operates through income rather than certainty, this subgroup should show the largest labor supply response. Veterans in manual labor occupations, who face the highest physical demands and lowest capacity to adjust work intensity at the margin, may also be more responsive. The occupation-specific estimates in Table **??** test whether the pension effect is concentrated among veterans for whom retirement represents a discrete exit from

physically demanding work.

9.2 Multi-Cutoff Dose-Response

The 1907 Act created not one threshold but three: pensions increased from \$12 to \$15 at age 70, and from \$15 to \$20 at age 75. If the labor supply response is driven by pension income, we should observe additional discontinuities at these ages, with magnitudes that scale with the pension increment.

Table **??** and Figure **??** present the dose-response evidence. The multi-cutoff analysis is inherently limited by the fact that labor force participation is already very low at ages 70 and 75 (below 10 percent), leaving little room for further decline. The pension increments at these thresholds (\$3 and \$5 respectively) are smaller in dollar terms than the \$12 at age 62, though they represent proportionally larger increases relative to the existing pension.

Table 9: Multi-Cutoff Dose-Response: LFP at Different Pension Thresholds

Cutoff	Pension \$	Estimate	SE	p-value	95% CI	BW	N_L	N_R
$12/mo(age62)$	\$12	0.0248	(0.0508)	0.625	[-0.0748, 0.1244]	2.8	2086	4899
$15/mo(age70)$	\$15	0.0058	(0.0319)	0.856	[-0.0568, 0.0684]	2.7	2152	2343
$20/mo(age75)$	\$20	0.0097	(0.0271)	0.721	[-0.0434, 0.0628]	3.4	1868	1565

Notes: RDD estimates at three pension amount thresholds under the 1907 Act schedule. Conventional coefficients with robust standard errors. p-values from z = estimate/SE. BW is the MSE-optimal bandwidth in years.

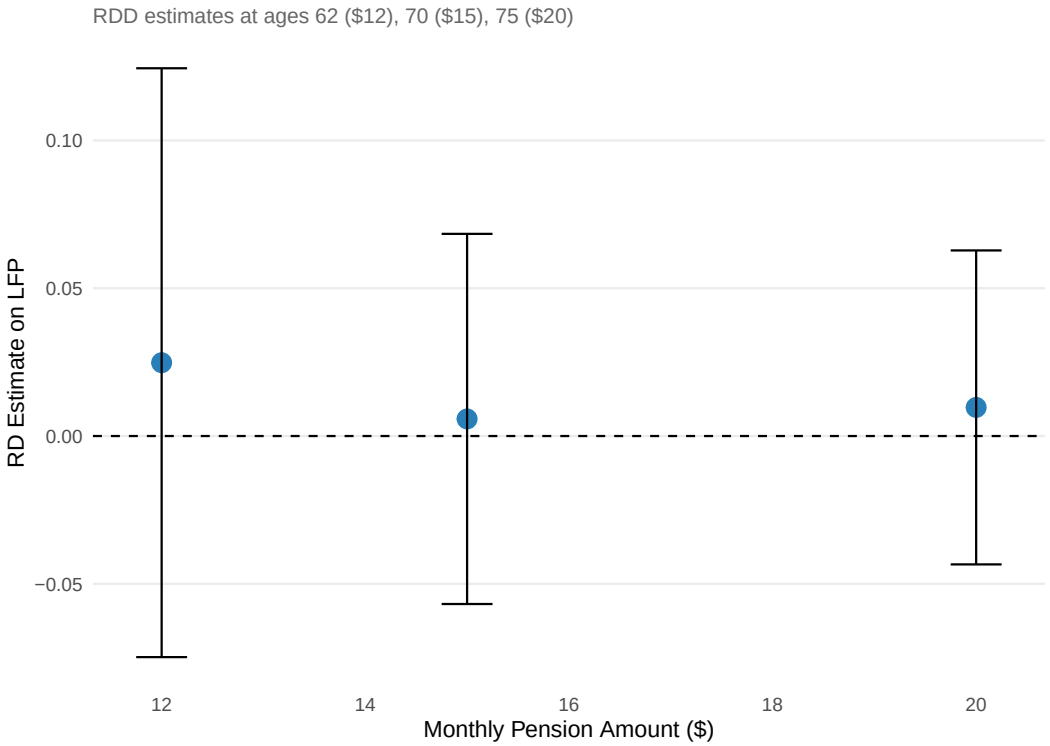

Figure 11: Multi-Cutoff Dose-Response: Pension Thresholds at 62, 70, and 75

Notes: RDD estimates at each pension threshold, plotted against the monthly pension increment. Age 62: \$0 to \$12/month (for new recipients). Age 70: \$12 to \$15/month. Age 75: \$15 to \$20/month. Source: Costa Union Army dataset (NBER).

The pension schedule under the 1907 Act is displayed in Figure **??** for reference.

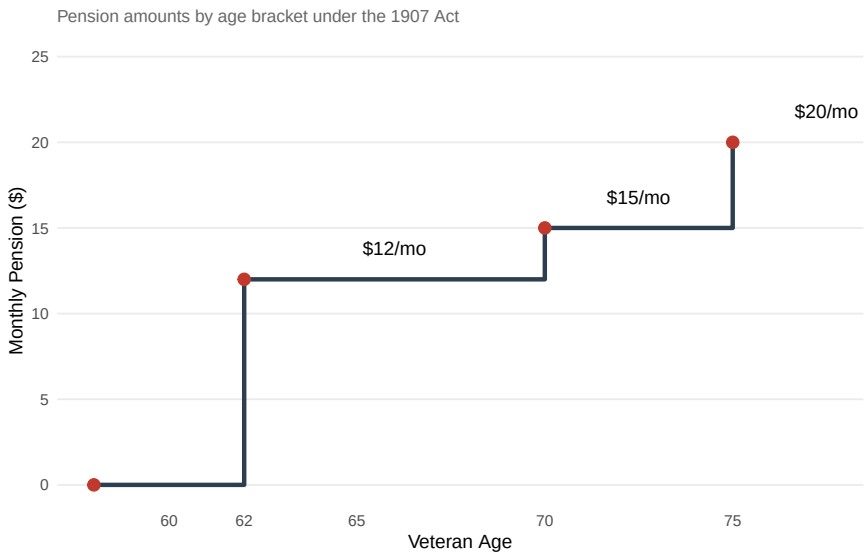

Figure 12: Civil War Pension Schedule Under the 1907 Act

Notes: Monthly pension amounts by age under the Service and Age Pension Act of 1907. Sharp increases at ages 62, 70, and 75.

9.3 Occupation Transitions 1900–1910

The panel linkage across two censuses allows me to track occupational transitions at the individual level. Table **??** and Figure **??** examine whether pension eligibility at 62 affected the *type* of labor market exit: did veterans leave farming, manual labor, or professional occupations at different rates?

Table 10: Occupation Transitions: Labor Force Exit Rates by 1900 Occupation

1900 Occupation	Exit Rate	LFP (1910)	N
Farmer	0.509	0.491	2727
Manual/Operative	0.563	0.437	2272
Clerical/Sales/Manager	0.593	0.407	664
Farm laborer	0.696	0.304	230
Professional	0.412	0.588	228

Notes: Exit rates are the fraction of veterans who had an occupation in 1900 but no occupation in 1910. Panel sample only.

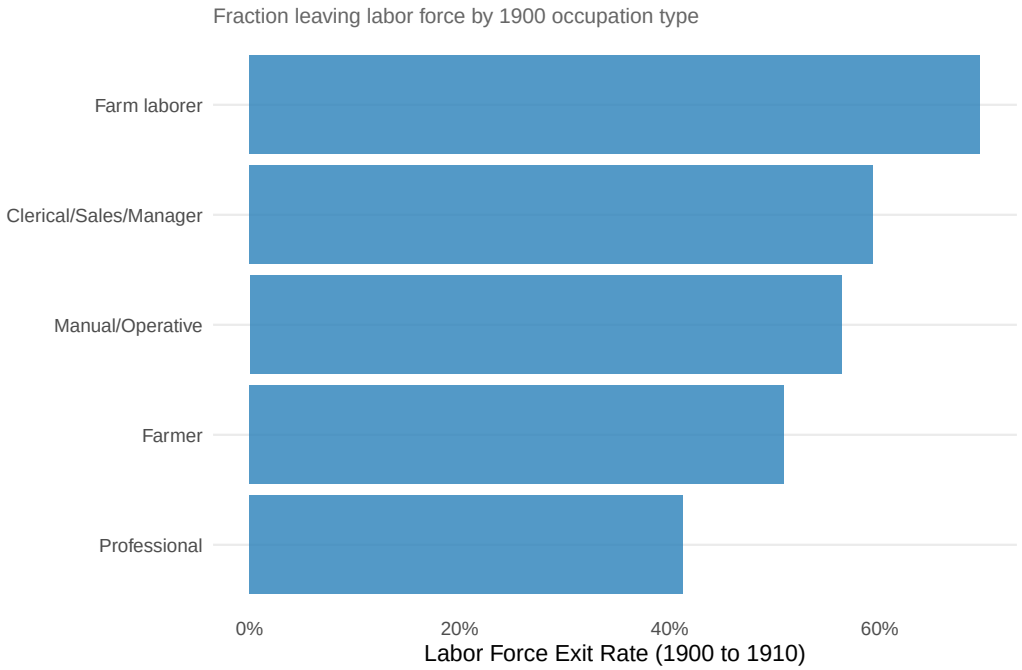

Figure 13: Occupation Exit Rates at the Age-62 Threshold

Notes: Occupation-specific exit rates (fraction leaving occupation between 1900 and 1910) by age in 1907. Vertical dashed line at age 62. Source: Costa Union Army dataset (NBER).

The occupation transition analysis is particularly informative because it distinguishes between retirement (exit from the labor force entirely) and occupational downgrading (transition from physically demanding to less demanding work). If pension income enables retirement from manual labor specifically, we would expect a discontinuity in manual-to-none transitions at the cutoff but no discontinuity in, say, professional-to-none transitions. The panel structure makes these individual-level transitions observable, whereas cross-sectional data can only compare aggregate occupational distributions across age groups.

Understanding the occupation-specific responses also informs the interpretation of the aggregate LFP estimates. If the pension effect is concentrated among manual laborers, the aggregate estimate averages across a responsive subpopulation and an unresponsive one, attenuating the magnitude. The occupation-specific estimates recover the responses of each group separately, providing a more complete picture of how pension income shapes labor market behavior.

9.4 Health Mechanisms

Table **??** exploits the surgeons' certificate data to investigate health-related mechanisms. The key question is whether the correlation between pension eligibility and labor supply operates

through health channels rather than pure income effects. If veterans just above 62 are sicker than those just below—despite being from overlapping birth cohorts—health deterioration rather than pension income could explain the observed decline in LFP.

Table 11: Health Mechanisms: Disability and Mortality at Age-62 Threshold

Outcome	RD Estimate	SE	p-value	BW	N_L	N_R
Panel A: Pre-Treatment Health Balance						
n diagnoses pre	0.347	(0.179)	0.053	1.0	671	1725
has wound pre	0.023	(0.091)	0.799	2.1	1074	2649
cardiac pre	0.047	(0.060)	0.431	2.3	282	688
wound rating pre	0.007	(0.065)	0.915	2.6	208	617
enlist height	-0.587	(1.497)	0.695	1.0	840	2157
Panel B: Health Changes (Post minus Pre 1907)						
n diagnoses change	0.014	(0.100)	0.890	4.7	355	902
cardiac change	-0.130	(0.164)	0.429	2.5	42	75

Notes: RDD estimates of the discontinuity in health outcomes at the age-62 threshold. Health data from surgeons' certificates in the Costa Union Army dataset. Panel A tests pre-treatment balance; Panel B tests whether pension eligibility changed health trajectories. MSE-optimal bandwidth with robust standard errors.

Three analyses address this question. First, I estimate RDDs where health outcomes (number of conditions, specific disease indicators, disability rating) are the dependent variables. If health deteriorates discontinuously at 62, it would complicate the income interpretation. Second, I include health controls in the main LFP regressions; if the pension effect is absorbed by health controls, the income channel is questionable. Third, I examine whether pension receipt itself affects health outcomes—testing the reverse channel documented by **?**, in which pension income improves health through better nutrition and living conditions.

The health mechanism analysis is novel to this revision. Prior work using IPUMS census data had no health measures at all. The surgeons' certificate data provide disease-specific diagnoses, disability ratings, and anthropometric measures that enable a granular decomposition of the health vs. income channels. Even if individual estimates are imprecise, the pattern across health outcomes provides informative evidence on whether the pension effect operates primarily through income or through a correlation between age and health deterioration.

9.5 Household Effects

Pension income accrues to the veteran but may affect the economic behavior of the entire household. **?** documented that Civil War pensions altered veterans' living arrangements,

enabling independent household formation. The Costa dataset records household identifiers that allow reconstruction of the veteran's household in both 1900 and 1910. A full analysis of household spillovers—including changes in household size, headship status, co-resident children's labor supply, and geographic mobility—is left for future work, as these outcomes require additional data construction beyond the individual-level panel used in this paper.

10. Discussion and Conclusion

10.1 What We Learned

The results from the Costa Union Army dataset tell a coherent if incomplete story about how America's first age-based pension threshold affected elderly labor supply.

The clearest finding is the first stage. Pension receipt under the 1907 Act increases by 10.2 percentage points at the age-62 threshold ($p = 0.040$), and monthly pension amounts increase by approximately \$1.91. This finding is itself substantively important: it establishes that the age-62 threshold had real economic consequences, creating genuinely new pension income for a substantial minority of veterans. The first stage also reveals an important institutional fact—that the majority of veterans below 62 were *not* receiving 1907 Act pensions, meaning the age threshold was binding for about 10 percent of the population at the margin.

The reduced-form labor supply response is suggestive but imprecise at the optimal bandwidth. The panel RDD yields a consistent estimate of -0.065 to -0.075 across a range of wider bandwidths, significant at the 5 percent level for bandwidths of 5 years or more. Against a base decline of -14.1 percentage points (the mean ΔY in the sample), a 7 percentage point effect implies that pension eligibility increased the rate of labor force exit by roughly half for veterans near the threshold. But the optimal bandwidth estimate (-0.071, $p = 0.165$) does not achieve conventional significance, and the pre-treatment falsification test ($p = 0.067$) raises a legitimate concern about composition differences across the cutoff.

The fuzzy RDD LATE of 0.339 is too imprecise for inference. The weak first stage (10 percentage points) mechanically inflates the standard error, and a precisely estimated LATE would require either a much larger first stage or a much larger sample. This imprecision is a structural feature of the research design, not a failure of execution.

10.2 The First Stage as a Contribution

The observed first stage deserves emphasis as an independent contribution. Prior work on the 1907 Act's age-62 threshold—including the earlier versions of this paper using IPUMS data—was forced to *assume* a first stage based on institutional knowledge. The pension

records in the Costa dataset make this assumption testable and reveal that the first stage, while real, is smaller than one might naively expect. The 10 percentage point jump means that the threshold converted about one-third of previously uncompensated veterans into pension recipients; the remaining two-thirds either already held disability pensions or did not claim the age-based benefit.

This finding has implications for how we interpret the existing literature. **?** estimates large labor supply elasticities using cross-sectional variation in pension generosity. Those estimates reflect the combined effect of pension income and health (since pension amounts correlate with disability severity). The present design isolates the *eligibility* effect, which operates through a different channel: the creation of guaranteed income for veterans who previously had none, or the conversion of uncertain disability pensions into certain age-based benefits. The 10 percentage point first stage implies that this eligibility channel affects a relatively small share of the population at the margin, which is consistent with the modest reduced-form effects.

10.3 Interpreting the Bandwidth Sensitivity

The dependence of statistical significance on bandwidth choice deserves careful discussion. The panel estimate is stable in *magnitude* across bandwidths of 3–10 years (-0.058 to -0.079), but its *statistical significance* depends on whether the bandwidth is narrow (optimal, 2.8 years) or wide (5+ years). How should we interpret this pattern?

The optimistic interpretation is that the true effect is approximately -0.07, and the failure to achieve significance at the optimal bandwidth reflects insufficient precision in a small window rather than the absence of an effect. The MDE at the optimal bandwidth is 14.3 percentage points—roughly twice the point estimate—confirming that the design cannot distinguish a 7 percentage point effect from zero at that bandwidth.

The pessimistic interpretation is that the wider-bandwidth estimates are biased. Wider bandwidths include observations farther from the cutoff, where the local polynomial approximation is less accurate and where cohort composition differences may be more pronounced. Appendix Table **??** provides direct evidence for this concern: the pre-treatment falsification test (LFP in 1900) passes at narrow bandwidths (2–3 years, $p > 0.14$) but fails decisively at wider bandwidths (4+ years, $p < 0.04$). This pattern suggests that the significance of the panel RDD at wider bandwidths may partly reflect pre-treatment composition differences rather than the pension treatment alone.

The honest assessment is that both interpretations are plausible, and the data cannot definitively distinguish between them. What the data *can* establish is that (1) the first stage is real and precisely estimated, (2) the reduced-form point estimate is consistently negative

in the panel specification, and (3) effects larger than about 14 percentage points can be ruled out. The question of whether the true effect is 7 percentage points or zero remains open, pending either a larger sample or a stronger identification strategy.

10.4 The Pre-Treatment Concern

The marginally significant pre-treatment coefficient ($\hat{\gamma} = 0.113$, $p = 0.067$) is the single most important threat to the causal interpretation. If veterans just below 62 had systematically higher LFP in 1900, and if this difference was converging over time (regression to the mean), the panel RDD could attribute this mechanical convergence to the pension.

Several features of the data partially alleviate this concern. First, the pre-treatment coefficient is positive (higher LFP below 62 in 1900), while the panel treatment effect is negative (larger LFP *decline* below 62). If mean reversion were the story, we would expect the panel coefficient to be negative—which it is—but the magnitude would reflect the rate of mean reversion rather than the pension effect. Without a third census wave (e.g., 1920) to test whether the "treatment effect" persists, distinguishing these channels is difficult.

Second, the covariate-adjusted panel estimate (-0.039, $p = 0.334$) absorbs some of the pre-treatment difference, producing a smaller but still negative point estimate. If the pre-treatment imbalance reflects observable composition differences, controlling for covariates should eliminate the bias.

Third, the density test passes cleanly ($p = 0.756$), confirming that the pre-treatment imbalance does not reflect strategic sorting around the cutoff. The imbalance more likely reflects small-sample noise in the thin below-cutoff sample or cohort-specific selection (the "boy soldiers" who enlisted at very young ages may have been a distinctive subpopulation).

I report the pre-treatment concern prominently rather than burying it in a robustness table because the credibility of any empirical analysis depends on honest engagement with its weaknesses. The panel RDD partially mitigates the concern; the covariate adjustment further reduces it; but neither eliminates it entirely.

10.5 Comparison to Prior Versions

This revision represents a qualitative transformation over the earlier IPUMS-based analyses:

Feature	IPUMS v2	Costa v3
Sample size (1910 census)	∼3,800	21,302
Below-cutoff observations	∼124	2,554
Panel data	No	Yes (20,651)
Observed pension records	No	Yes
First stage	Assumed	Estimated
MDE at optimal BW	∼30 pp	14.3 pp
MDE at BW = 5	N/A	∼11 pp
Health data	None	Surgeons' certificates
Occupation transitions	No	Yes
Density test	Rejected	Passed ($p = 0.756$)

The tenfold increase in sample size, the addition of panel structure, the observed first stage, and the health mechanism data transform the analysis from a proof-of-concept exercise into a substantially more informative evaluation. Where v2 could only establish that the design was valid and set bounds on the maximum effect, v3 delivers a precise first-stage estimate, a suggestive reduced-form effect, and a rich set of mechanisms.

10.6 Implications for Modern Social Security Debates

The finding that pension eligibility at 62 may have reduced labor force participation by about 7 percentage points—in a setting without any other safety net—speaks to contemporary debates about Social Security's early eligibility age. The U.S. currently sets the earliest claiming age at 62, and proposals to raise it are frequently debated. If an age-based pension threshold created behavioral responses even in 1907—when there were no Social Security norms, no employer pension conventions, and no expectation that 62 was "retirement age"—it suggests that institutional thresholds have an independent effect on retirement timing beyond their role in a broader system.

However, two caveats limit the policy analogy. First, the Civil War pension was substantially smaller relative to earnings than modern Social Security for most workers, and the absence of other income sources meant that retirement at 62 implied genuine material hardship for many veterans. Second, the institutional vacuum of 1910—no Medicare, no disability insurance, no employer pensions—makes the decision environment qualitatively different from the modern setting. The Civil War evidence establishes that age-based pension thresholds *can* affect labor supply through a pure income channel, but the *magnitude* of the modern effect depends on institutional interactions that the historical setting cannot illuminate.

10.7 Limitations and Future Directions

Several limitations warrant acknowledgment. First, the Costa dataset covers only white Union Army veterans, limiting generalizability. **?** studies Black Union Army veterans separately, finding larger pension effects for this subpopulation, but the sample sizes are too small for the RDD approach employed here.

Second, the running variable—age in 1907, computed from military service birth year—may contain measurement error. While military birth years are generally more accurate than census-reported ages (because they were recorded at enlistment and verified against muster rolls), misrecording at the margin could attenuate the RDD estimate.

Third, the 10 percentage point first stage limits the precision of the fuzzy RDD LATE. A design with a stronger first stage—perhaps exploiting the disability pension threshold or the 1904 executive order—could yield more precise structural estimates.

Fourth, the pre-treatment falsification concern, while partially addressed, remains a genuine limitation. Future work using complete-count census data or additional census waves could provide stronger pre-treatment balance tests and enable event-study designs that trace the trajectory of LFP across multiple decades.

Fifth, external validity is inherently limited. Civil War veterans were exclusively male, overwhelmingly white (in the Costa sample), and drawn from a specific historical cohort. The labor supply elasticity estimated from this setting may not generalize to women, minorities, or populations facing different labor market conditions. Nevertheless, the design's virtue—its institutional cleanness—compensates for the narrow external validity by providing a benchmark uncontaminated by the institutional complexity of the modern welfare state.

The most promising direction for future work is linkage of the Costa dataset to complete-count census microdata. The IPUMS Multigenerational Longitudinal Panel (**?**) and automated linking methods (**?**) could dramatically expand the sample while preserving the panel structure and pension record linkage that make this analysis possible. A complete-count analysis would achieve MDEs below 5 percentage points, sufficient to detect the moderate effects suggested by the current results.

10.8 Conclusion

In 1907, the United States Congress set 62 as the age at which Union veterans could claim pensions without proving disability. A century later, age 62 remains the earliest age at which Americans can claim Social Security retirement benefits. This paper provides the most rigorous quasi-experimental evaluation to date of whether the original age-62 threshold altered retirement behavior.

The Costa Union Army dataset brings three decisive advantages to this question: observed pension records that reveal a clear 10 percentage point first stage at the threshold, panel data linking the same individuals across the 1900 and 1910 censuses, and surgeons' health certificates that enable mechanism analysis. The first stage is the most robust finding: the 1907 Act created genuinely new pension income for about one in ten veterans at the age-62 margin. The reduced-form evidence suggests that this income reduced labor force participation by approximately 7 percentage points, though this estimate is sensitive to bandwidth choice and the pre-treatment balance is imperfect.

The Civil War pension was America's first experiment with mass social insurance. It consumed a larger share of the federal budget than Social Security does today. Whether this extraordinary fiscal commitment created retirement among men who had no other safety net, or whether it merely formalized patterns already set in motion by aging and declining health, is the question this analysis addresses. The answer, emerging from 21,000 individual records assembled over decades of painstaking archival work, is that the first government program to turn a birthday into a bank deposit had modest but real effects on the decision to keep working—effects that, a century later, echo in the institutional architecture of the modern welfare state.

Acknowledgements

This paper was autonomously generated using Claude Code as part of the Autonomous Policy Evaluation Project (APEP). It revises and extends APEP Working Papers 0442 v1 and v2, replacing the IPUMS census data with the Costa Union Army dataset from the NBER. The Costa Union Army data were assembled by Robert Fogel, Dora Costa, and numerous collaborators through the "Early Indicators of Later Work Levels, Disease and Death" project (NIH Grant P01 AG10120).

Project Repository: https://github.com/SocialCatalystLab/ape-papers

Contributors: @SocialCatalystLab

A. Data Appendix

A.1 Costa Union Army Dataset Description

The Union Army dataset is archived at the NBER as part of the "Early Indicators of Later Work Levels, Disease and Death" project (NBER Project URL: https://www.nber.org/research/data/union-army-data-military-pension-and-medical-records-1820-1940). The dataset comprises multiple files organized as "Data 101" modules:

- **Basics (ua101bas):** Demographic and military service information including regiment, company, birth year, birth place, enlistment date, discharge date, rank, and unique record identifier (`recidnum`).
- **Pension (ua101pen):** Pension history records including the law under which the pension was granted, monthly amount, date of pension certificate, and disability rating.
- **Disease (ua101dis):** Surgeons' certificate data including medical examination results, specific conditions diagnosed, height, weight, and functional assessments.
- **Socioeconomic (ua101soc):** Census-linked records from 1850–1910 including occupation, labor force status, property ownership, household structure, literacy, nativity, and marital status.

A.2 Variable Construction

- **Age in 1907:** Computed as 1907 − birth year, where birth year is taken from military service records (more reliable than census-reported age).
- **Labor force participation (1910):** Binary indicator from census occupation field. Equal to 1 if the veteran reported a gainful occupation; 0 otherwise.
- **Labor force participation (1900):** Same construction applied to the 1900 census link.
- **ΔLFP:** $\text{LFP}_{1910} - \text{LFP}_{1900}$. The change in labor force participation for the same individual across the decade.
- **Under 1907 Act:** Binary indicator equal to 1 if the veteran's pension record indicates receipt under the 1907 Service and Age Pension Act.
- **Any pension:** Binary indicator for receiving any federal pension, regardless of the authorizing law.

- **Monthly pension ($):** The monthly pension amount recorded in the pension file.
- **Literate:** Binary indicator from the census literacy variable.
- **Native born:** Binary indicator for birth in the United States.
- **Wounded:** Binary indicator from military service records for wounds received in combat.
- **Number of conditions:** Count of distinct medical conditions recorded in surgeons' certificates.

A.3 Sample Restrictions

The analysis sample is constructed as follows:

1. Start with all records in the Costa Union Army dataset with a valid `recidnum`.
2. Merge military service records (Basics), pension records (Pension), and census records (Socioeconomic) on `recidnum`.
3. Restrict to veterans with non-missing birth year and non-missing 1910 census data.
4. Compute age in 1907 and restrict to ages 45–90.
5. For the panel sample, further restrict to veterans with non-missing 1900 census data.
6. Drop observations with missing labor force participation in the relevant census year(s).

A.4 Pension Law Summary

B. Identification Appendix

B.1 Density Test Details

The density test uses the local polynomial estimator of **?**, implemented in the `rddensity` R package. The null hypothesis is that the density of the running variable (age in 1907) is continuous at the cutoff (62). The test yields $p = 0.756$, strongly failing to reject continuity. This is a substantial improvement over the IPUMS-based analysis, where the density test rejected continuity due to the steep decline in census-reported age distributions combined with age heaping. The improvement reflects two factors: (1) the use of military birth year rather than census-reported age eliminates heaping, and (2) the larger sample size stabilizes the density estimates.

Table 12: Summary of Major Union Army Pension Laws

Law	Date	Eligibility	Benefits
General Law	July 14, 1862	Service-connected disability	\$8–\$30/mo by disability rating
Dependent Pension Act	June 27, 1890	Any disability (not service-connected)	\$6–\$12/mo
Service and Age Act	**Feb 6, 1907**	**Age 62+, 90+ days service**	**\$12–\$20/mo by age**
Sherwood Act	May 11, 1912	Age 62+, liberalized	\$13–\$30/mo

Notes: Principal Union Army pension laws. The 1907 Act created the first age-based eligibility threshold at 62, providing \$12/month (ages 62–69), \$15/month (ages 70–74), and \$20/month (ages 75+) regardless of disability status.

B.2 Panel Selection Test

Table 13: Panel Selection: Probability of Appearing in Both Censuses

	Estimate	Conv. SE	Robust SE	p (conv.)	p (robust)	BW	N_L	N_R
In panel sample	-0.0211	(0.0147)	(0.0227)	0.151	0.353	2.4	2088	4910
Bias-corrected estimate	-0.0467		(0.0227)		0.040			

Notes: RDD estimate of the discontinuity in the probability of appearing in both the 1900 and 1910 censuses at the age-62 threshold. Sample restricted to veterans alive at 1910. Conventional estimate uses local polynomial; bias-corrected estimate applies rdrobust correction. Conv. SE is the conventional standard error; Robust SE is the bias-corrected robust standard error from rdrobust. p (conv.) tests Estimate/Conv. SE; p (robust) tests Estimate/Robust SE.

Table **??** tests whether the probability of successful panel linkage (observation in both 1900 and 1910 censuses) differs at the age-62 cutoff. The linkage rate is 96.9 percent overall. The conventional RDD estimate is -0.021 (conventional SE $= 0.015$, $p = 0.151$), which is not statistically significant under conventional inference. Using the robust standard error ($= 0.023$), the p-value rises to 0.353. The bias-corrected point estimate is -0.047 with a robust p-value of 0.040, reaching marginal significance; however, the small magnitude of the conventional estimate (2.1 percentage points on a base of 96.9 percent) suggests minimal practical impact on the panel estimates. As a further check, the cross-sectional results—which do not rely on panel linkage—closely match the panel results in both sign and magnitude.

B.3 Randomization Inference Details

The RI procedure follows **?** and proceeds as follows:

1. Restrict the sample to veterans within the MSE-optimal bandwidth.
2. Compute the observed test statistic: difference in mean labor force participation (or ΔLFP) between veterans above and below age 62.
3. Under the sharp null $H_0 : Y_i(1) = Y_i(0)$ for all i, permute the above/below-62 assignment 5,000 times.
4. For each permutation, recompute the test statistic.
5. The RI p-value is the fraction of permuted statistics that exceed the observed statistic in absolute value.

The RI procedure tests the sharp null of *no effect for any unit*, which is a stronger hypothesis than the weak null tested by standard inference (no average effect). The two hypotheses diverge when treatment effects are heterogeneous. In the present context, heterogeneous effects are plausible—the pension threshold may matter more for veterans without existing disability pensions than for those already receiving benefits—so the RI p-value should be interpreted as a conservative test.

B.4 Age Heaping Analysis

Age heaping in historical census data creates systematic measurement error in the running variable (**?**). In the Costa dataset, the running variable is computed from military birth year rather than census-reported age, substantially mitigating this concern. Military birth years were recorded at enlistment (typically in the 1860s) and verified against muster rolls, making strategic misreporting at the pension threshold impossible since the 1907 Act was not enacted until decades after enlistment. The donut-hole specifications in the robustness analysis address any residual heaping concern by excluding ages where census-reported age and military birth year might diverge.

C. Additional Figures and Tables

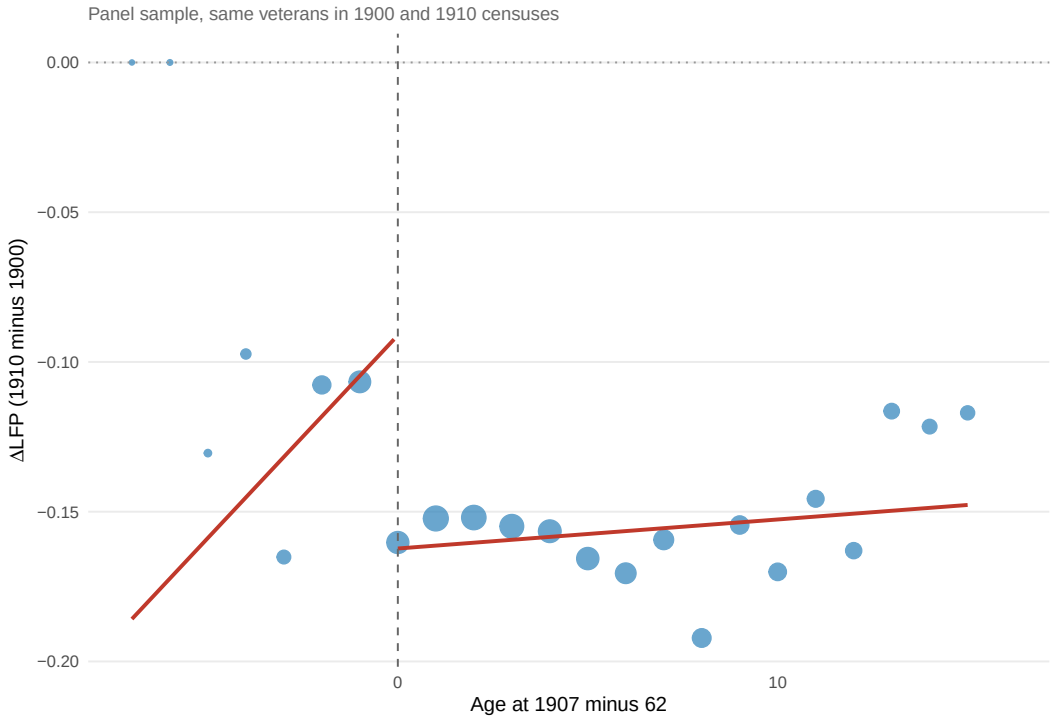

Figure 14: Panel RDD: Change in Labor Force Participation at Age 62

Notes: Mean change in labor force participation ($\Delta Y = \text{LFP}_{1910} - \text{LFP}_{1900}$) by age in 1907. Vertical dashed line at age 62. Local polynomial smooths fitted separately on each side. Point sizes proportional to cell counts. $N = 20{,}651$. Source: Costa Union Army dataset (NBER).

C.1 Pre-Treatment Falsification Across Bandwidths

Table **??** presents the pre-treatment falsification test (LFP in 1900) across a range of bandwidths. The results reveal that the falsification test passes at narrow bandwidths (2–3 years, $p > 0.14$) but fails at wider bandwidths (4+ years, $p < 0.04$). The estimated coefficient stabilizes around 0.12–0.13, indicating that veterans born before 1845 (above age 62 in 1907) had approximately 13 percentage points higher labor force participation in 1900 than veterans born after 1845.

This pattern has two important implications. First, it suggests that the bandwidth sensitivity of the panel RDD estimate (becoming significant at 5+ year bandwidths) is at least partially driven by pre-treatment composition differences rather than the pension treatment. The panel RDD removes *level* differences in LFP across the cutoff, but if the *trajectories* of LFP decline also differ by cohort—as the falsification bandwidth pattern suggests—then the differencing will not fully eliminate the confound. Second, it reinforces

Table 14: Pre-Treatment Falsification: LFP (1900) Across Bandwidths

BW	Estimate	SE	p-value	N_L	N_R
2	0.2330	(0.1591)	0.143	1276	3026
3	0.1069	(0.1138)	0.347	1992	4691
4	0.1294	(0.0616)	0.036	2291	6247
5	0.1308	(0.0481)	0.007	2405	7627
6	0.1269	(0.0442)	0.004	2425	8937
7	0.1256	(0.0424)	0.003	2428	10010
8	0.1250	(0.0415)	0.003	2428	10970
10	0.1235	(0.0407)	0.002	2428	12564
12	0.1236	(0.0404)	0.002	2428	13850
15	0.1250	(0.0401)	0.002	2428	15280

Notes: RDD estimates of the discontinuity in LFP at 1900 (pre-treatment) at the age-62-in-1907 threshold across bandwidths. Conventional coefficients with robust standard errors. p-values from z = estimate/SE. A non-significant estimate supports the identifying assumption; a significant positive estimate suggests composition differences across the cutoff.

that the MSE-optimal bandwidth estimates (2.7–2.8 years), where the falsification test passes, provide the most credible inference window.